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MAGNETOSTRATIGRAPHY AND GEOCHRONOLOGY OF NORTHWEST ICELAND

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Abstract. The NW peninsula of Iceland is built mainly by tholeiitic, low K flood basalts that show a regional southeasterly dip, generally of less than 10°. From the lavas mapped and drilled for paleomagnetic measurements (a total of 1261 flows), two continuous composite sections were constructed, one on the western side of the peninsula 4055 m thick comprising 456 lavas and the other adjacent to the east coast, 3165 m thick comprising 396 lavas. K-Ar age measurements were made on more than 70 lavas distributed throughout the two sections. These two lava sequences are about 90 km apart along strike, and although direct stratigraphic correlation is not possible between them, the evidence indicates that the eastern section is stratigraphically younger than that in the west, with some overlap. The results show that the lava sequences were erupted over an interval extending from about 14 to 8 Ma ago in the middle to late Miocene. Remarkably linear relations are found between the measured age and cumulative stratigraphic thickness for both sections, indicating rather uniform local rates of growth of the lava piles. For the western composite section a growth rate of 1820 m/Ma is found, contrasting with a much lower growth rate of 670 m/Ma for the eastern sequence. The average thickness of individual lavas in both sections is close to 8 m. The average time between eruptions of lavas in the western section is 5000 years, compared with a value of 12,000 years for the eastern section. The regional geology together with our measurements are interpreted in terms of crustal accretion processes by which the NW peninsula of Iceland was constructed by eruption of lavas from a spreading center of northeasterly strike. It is not clear whether the differences in rates of growth of the lava pile in the western and eastern sections are because of variation in eruptive activity with time or because eruptive activity has varied along the strike of the spreading center. Paleomagnetic measurements on the lavas provide a detailed history of polarity of the geomagnetic field during the interval 14-8 Ma ago. For the eastern composite section there is excellent and unambiguous correlation of the observed polarity pattern with the marine magnetic anomaly record. From these data we obtain an age of 9.6 Ma for the younger boundary of marine magnetic anomaly 5 and derive a revised polarity time scale for part of the Miocene which gives ages for individual polarity interval boundaries that are more than 5% greater than in recently published polarity time scales. In the revised time scale anomaly 5 time extends from 9.6 to 11.1 Ma ago.

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Introduction

Iceland lies astride the Mid-Atlantic Ridge and has developed essentially through voluminous basaltic magmatism. Since the work of Bödvarsson and Walker [1964] it has become generally accepted that Iceland has grown by crustal accretion through volcanism along the axial rift zone, which, in terms of the plate tectonic paradigm, marks the boundary between the North American and Eurasian plates. Consistent with this view, in general the lavas become progressively older away from the axial rift zone, although in detail there is evidence that the spreading zone has shifted laterally at various times in the past [Palmason and Saemundsson, 1974]. Thus there are nearly continuous sequences of flood basalts, erupted subaerially, which range in age from the present back to about 16 Ma or so [Moorbath et al., 1968], and these oldest lavas occur adjacent to the east and west coasts of Iceland.

In previous studies of the extensive lava sequences in Iceland we have shown that through stratigraphic mapping of the flood basalts combined with paleomagnetic and K-Ar age measurements, direct information on many aspects of the history of the geomagnetic field, including the time scale of polarity reversals, can be obtained [McDougall et al.,1977; Watkins et al., 1977; Saemundsson et al., 1980; Kristjansson and McDougall, 1982]. For times earlier than about 5 Ma ago it is essential to use stratigraphic sequences in refining the polarity time scale because of the decreasing resolution of the K-Ar dating method with increasing age. Extension of the polarity time scale by direct measurement is of some importance, as the time scales in use are based essentially upon extrapolations from land based K-Ar and magnetic polarity data for the last 3.5 Ma, combined with analysis of linear marine magnetic anomalies.

In this study we present results from the NW peninsula of Iceland, known to be one of the oldest exposed parts of the lava pile. Stratigraphic mapping of the lava sequence, exceeding 7 km in total thickness, together with detailed paleomagnetic and K-Ar age measurements are used to document the history of the geomag-

netic field over some 6 Ma of late Tertiary time. In addition, our studies provide much basic information relevant to the geological history of Iceland.

This work forms part of a major project on Icelandic lava sequences undertaken through the initiative of, and in collaboration with, the late Norman D. Watkins, University of Rhode Island.

Geology

Previous Geological Studies

The main features of the geology of the NW peninsula have been known since Thoroddsen [1906]. He noted a consistent southeasterly dip of the basalts and assumed that lignite beds found throughout the area belonged to a single stratigraphic level. Large faults had to be postulated to allow such a correlation. In this way he avoided the problem of a great cumulative thickness of the lava succession. Einarsson [1960] published a map of the regional dips over the entire region. By that time the idea of many lignite beds at different stratigraphic levels was gaining support, and the great thickness of the lava succession was obvious [Einarsson, 1963]. The first K-Ar dating studies on rocks from the NW peninsula were made on samples from near the town of Isafjördur, following detailed mapping of the local geology [Sigurdsson and Einarsson, unpublished manuscript, 1967]. The samples were obtained at high altitude above the zeolite alteration zones, and the measured ages averaged 16.0 ± 0.3 Ma [Moorbath et al., 1968], truly a sensationally low age at that time, which significantly changed the views about the genesis of Iceland. Reevaluation of these data and new measurements from this locality yielded an average age of 14.1 ± 0.2 Ma [Albertsson and Einarsson, 1982].

Mapping

Selective mapping was done on lava successions as remote as possible from central volcanoes, so as to minimize inclusion in the overall section of thick sequences erupted rapidly from such volcanoes. We ended up with essentially two long main composite sections trending N-S obliquely across the regional strike of the lava pile; one adjacent to the west coast of the peninsula, the other adjacent to the east coast (Figure 1). The distance between them is about 90 km measured along strike. A short segment from the intervening region probably overlaps with both main sections (Figure 1). The western composite section is 4055 m thick, consisting of 456 lavas, and the eastern composite section is 3165 m thick, comprising 396 lavas (Figure 2).

The mapping did not extend far beyond the sections apart from what was considered necessary to establish a reliable stratigraphic tie from one profile to another. For control of the general structural features the major sedimentary beds were traced in the field and the regional dips determined. This helped define such features as thickness variations along strike, regional flexures, and discontinuities in the stratigraphic succession. Because of the low relief in some parts of the area a large number of profiles was necessary to put together a continuous section. The ties were based variously on lithologic criteria and paleomagnetic sign determined by field measurement. Only distinctive tuff layers enabled unambiguous correlations; elsewhere there is a factor of uncertainty involved, and a few flows may have been double counted or missed out. However, this does not significantly upset the overall continuity in the stratigraphic record.

Rock Types

The lava types are similar to those mapped in eastern Iceland by Walker [1959]. The approximate frequency of rock types as determined for the western section, the intervening section (SF+SZ), and the lowest 1100 m of the eastern section, altogether 6300 m of strata, was found to be tholeiite 51%, olivine tholeiite 16%, porphyritic basalt 29%, and intermediate and acid lavas <0.5%. Interbeds including red baked soil, basaltic tuff, ash layers, and redeposited detritus constitute about 4% of the total stratigraphic column. Among the porphyritic lavas (more than 5% phenocrysts), big feldspar phenocrysts are common and sometimes very abundant. We have marked these lavas as of cumulate type in the sections as the plagioclase is of bytownite-anorthite composition.

Central volcano successions are intercalated with the composite sections locally, being distinctive as thin tholeiite lavas together with some intermediate to acid flows. The upper 2700 m of the eastern section was determined to be 85% tholeiite; this predominance of tholeiite may be because of closeness to central volcanoes, perhaps borne out by the presence of rather thin flows in this part of the section.

The lava succession, with its generally thin interbeds of eolian origin, contains a number of thicker detrital beds. They are composed mainly of basaltic ash, as are the thin red interbeds. Assuming constant eolian accumulation rate with time, they may represent long quiet intervals. Acid tuffs, hyaloclastite, conglomerate, mudstone, and lignite also occur within them. From the geology, however, there is only one significant hiatus which occurs at the lowest thick interbed, some 300 m above the base of the exposed section. This conclusion is reached because of the extensive weathering of the basalts below. This interbasaltic bed is mappable over 60 km along strike with clayey soil, limonite crusts, tuffs, and lignite. Several other interbeds traceable as marker beds or groups over long distances lack a comparable weathering of the substratum and are therefore regarded as marking shorter intervals. They may, however, indicate lulls in the volcanic production rate perhaps because new volcanic systems become active farther to the SE than previously.

Secondary Mineralization

Secondary mineralization of the lava pile is a corollary of its growth. Subsidence keeps pace with the extrusion of lavas, and the pile founders into a higher-temperature regime, where the mineralogy of the basalt starts to break down to

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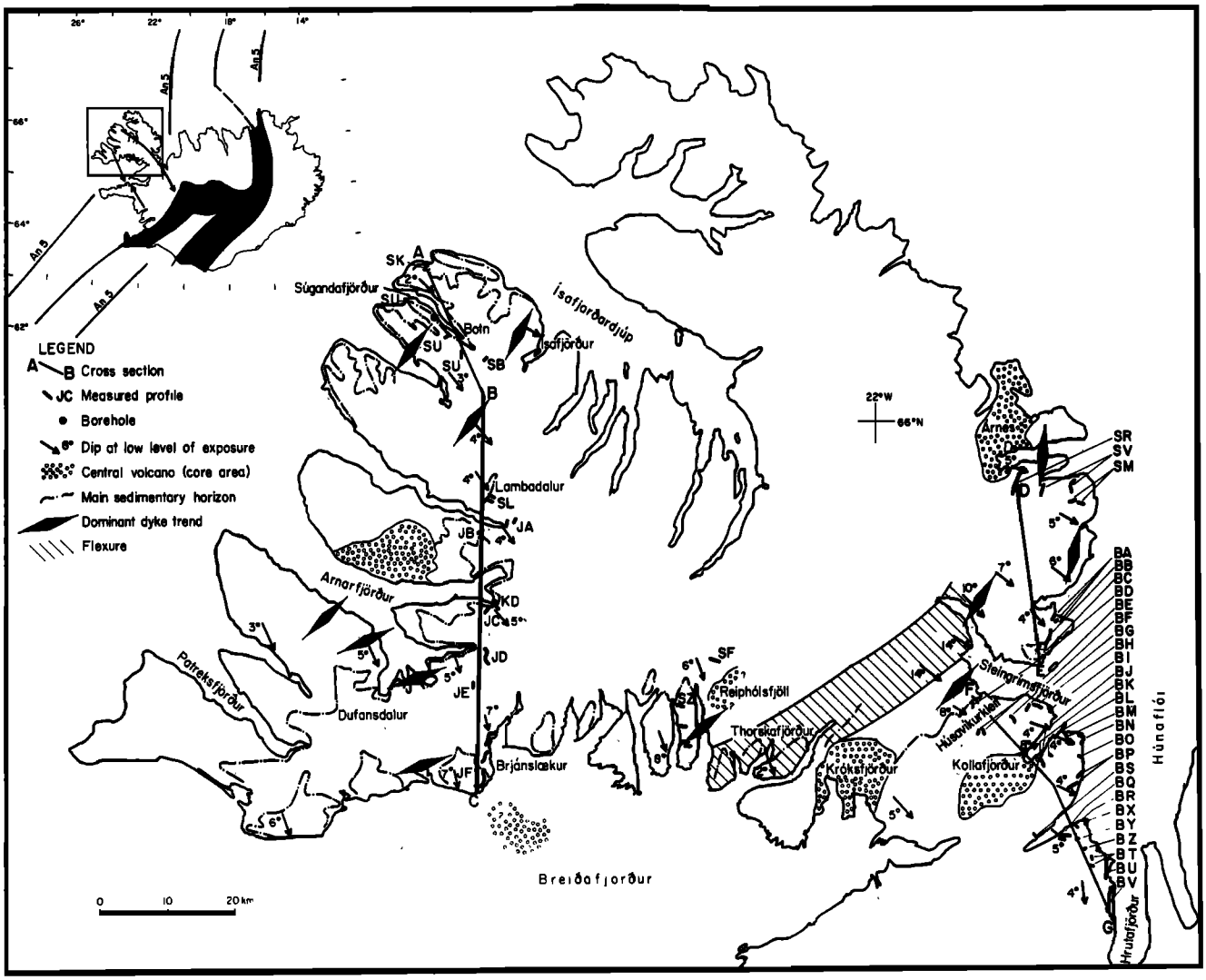


Fig. 1. The NW peninsula of Iceland, showing the main structural features and location of profiles which together comprise the western and eastern composite sections. Inset shows location of the peninsula relative to the presently active volcanic zones in Iceland, shown in black. Regional dips are indicated by arrows. The position of anomaly 5 from marine magnetic surveys is shown.

form secondary minerals, notably silica minerals, zeolites, and clay. Mineralogical zoning with depth was established for the lava pile in eastern Iceland [Walker, 1960]. Similar zoning is present in the NW peninsula. However, the lava pile is less deeply eroded so that the deeper regional zeolite zones are not exposed in the NW peninsula. The degree of alteration corresponds to temperatures up to about 100°C at the deepest levels of exposure [Kristmannsdóttir, 1979]. Hydrothermal aureoles, where higher temperatures prevailed, were observed around intrusive centers of central volcanoes. Figure 3 shows the position of the profiles in relation to the zeolite zones. The western composite section is generally at a higher level of exposure than the eastern one. This is mainly because of the generally low topography on the east side of the peninsula, particularly south of Steingrimsfjördur. No zeolite zones are indicated south of Steingrimsfjördur because of the predominance of tholeiite lavas and because the low topography made their definition difficult. However, it would appear that the top of the mesolite-scolecite zone is well above sea level.

Structure

The exposed rocks of the NW peninsula are the uppermost 1 km of a lava plateau that is primarily extrusive to at least 4-5 km below sea level [Flóvenz, 1980]. The general structure is rather simple as basalt lavas dip mainly at low angles toward the SSE through ESE, indicating a structural dome with its apex at the entrance of Isafjardardjúp. The change in strike also involves dikes and faults and is a regional feature of the geology of Iceland through time [Saemundsson, 1978]. Steeper dips up to 10°-15° define a flexure running from the inner part of Breidafjördur to Steingrimsfjördur (Figure 1). This could be related to vigorous

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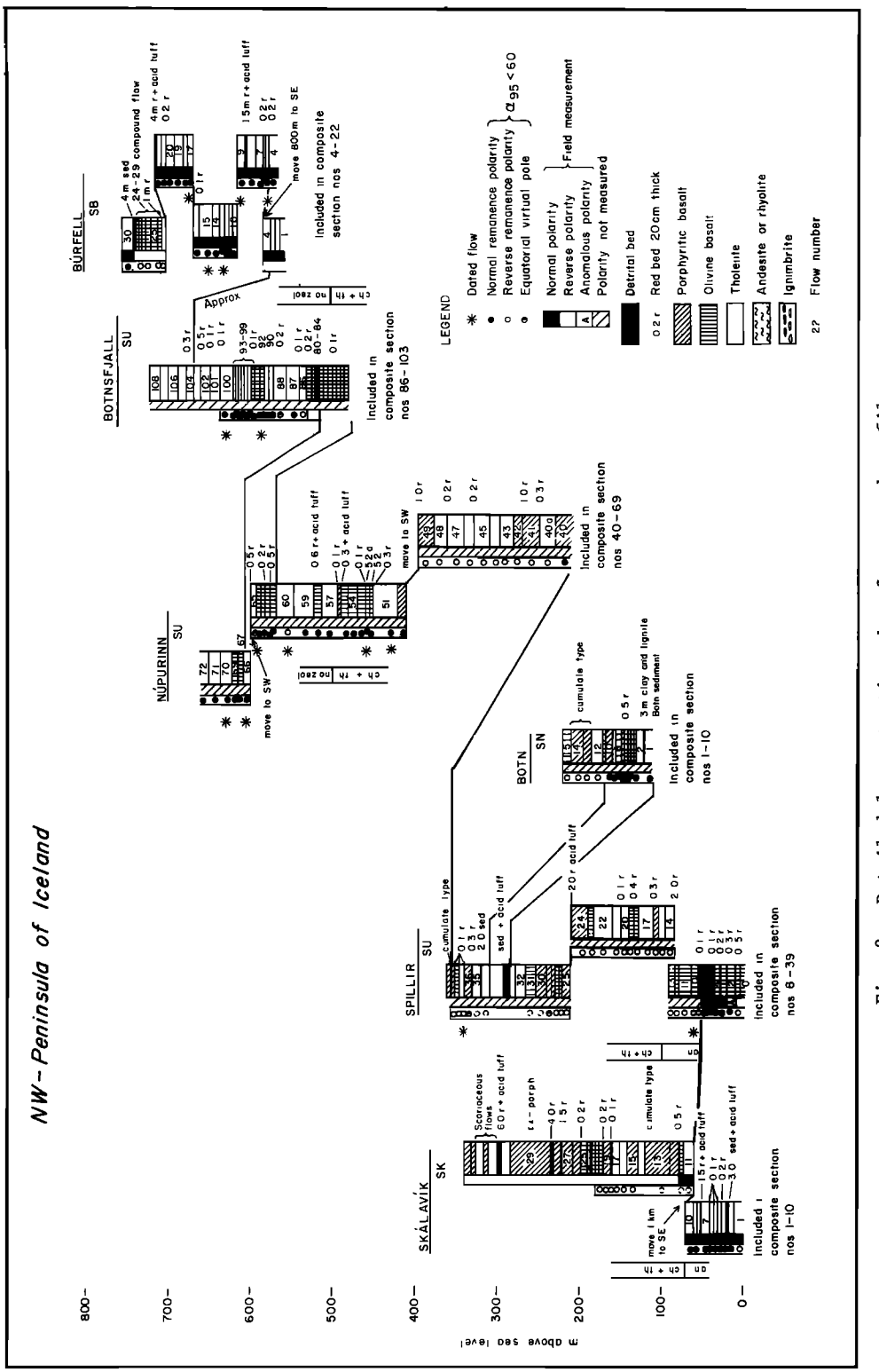


Fig. 2. Detailed lava stratigraphy of measured profiles.

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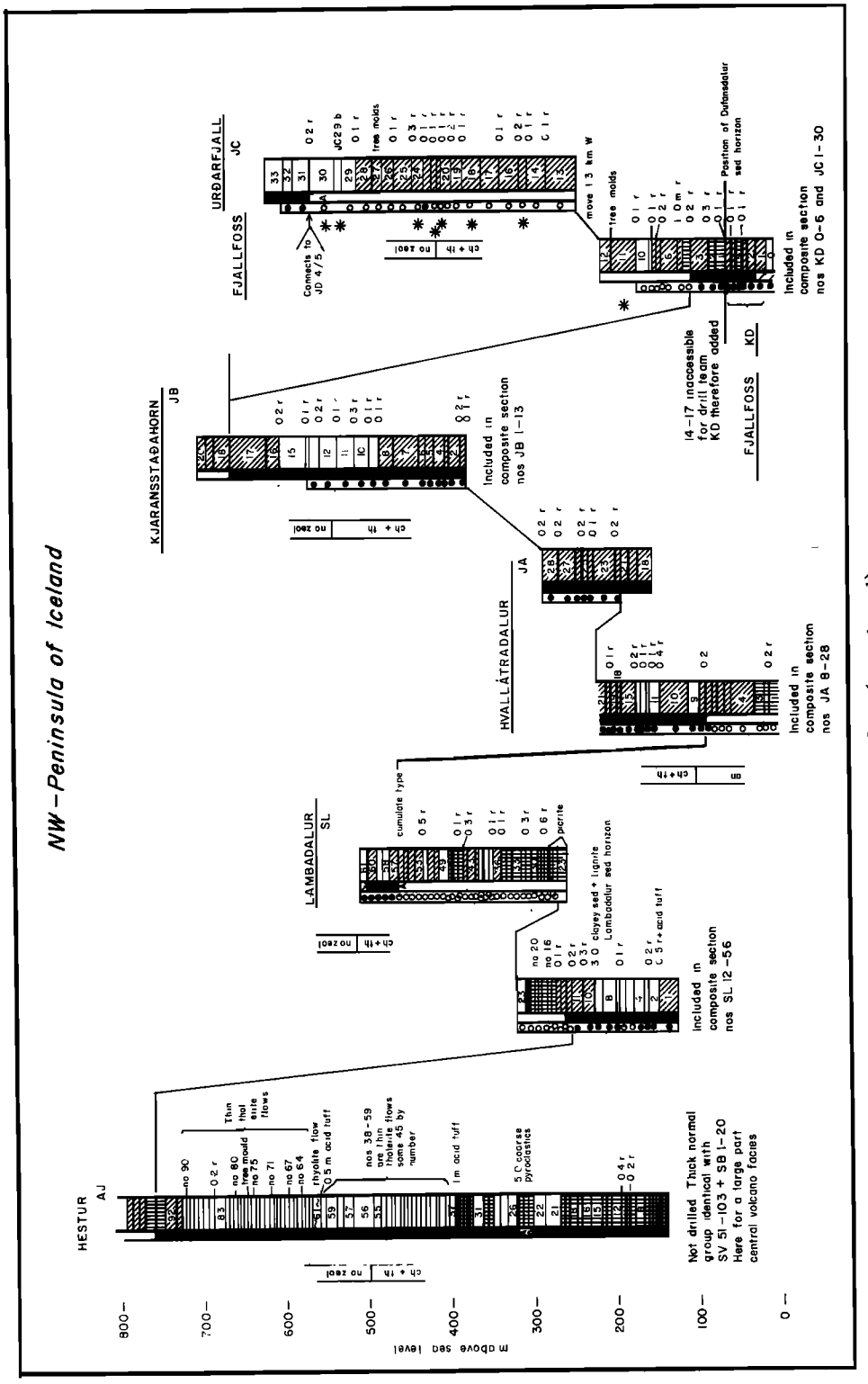


Fig. 2. (continued)

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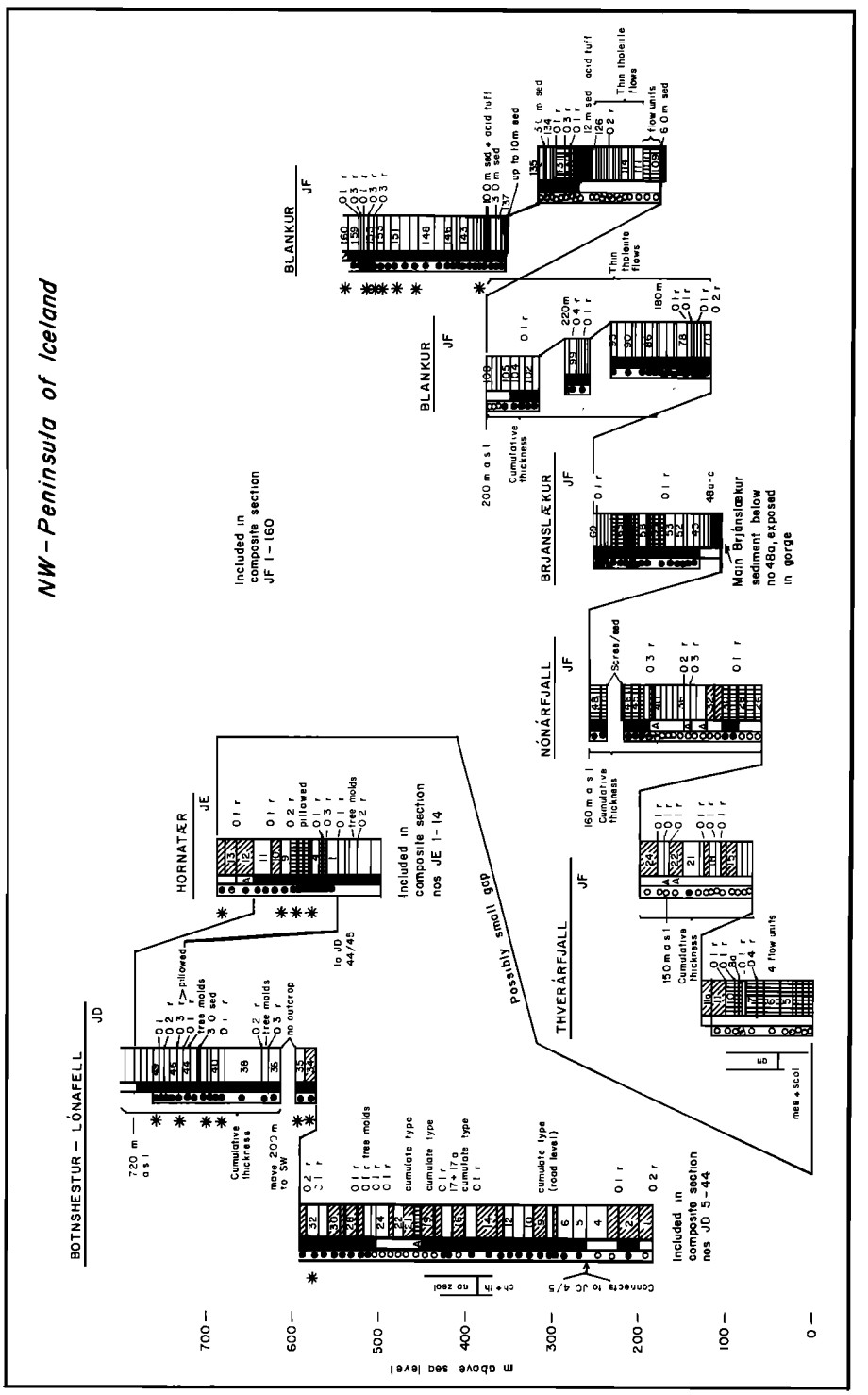


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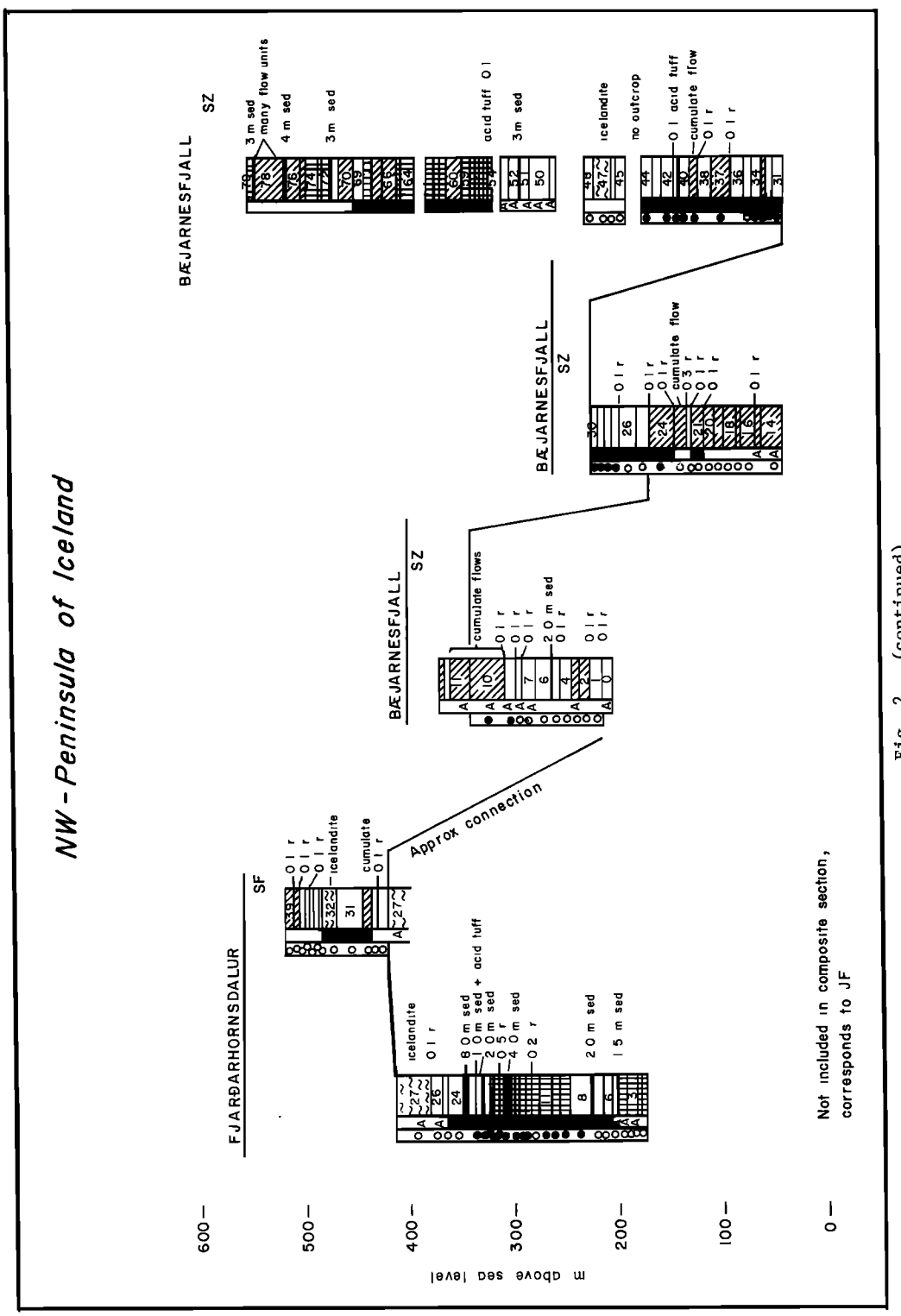


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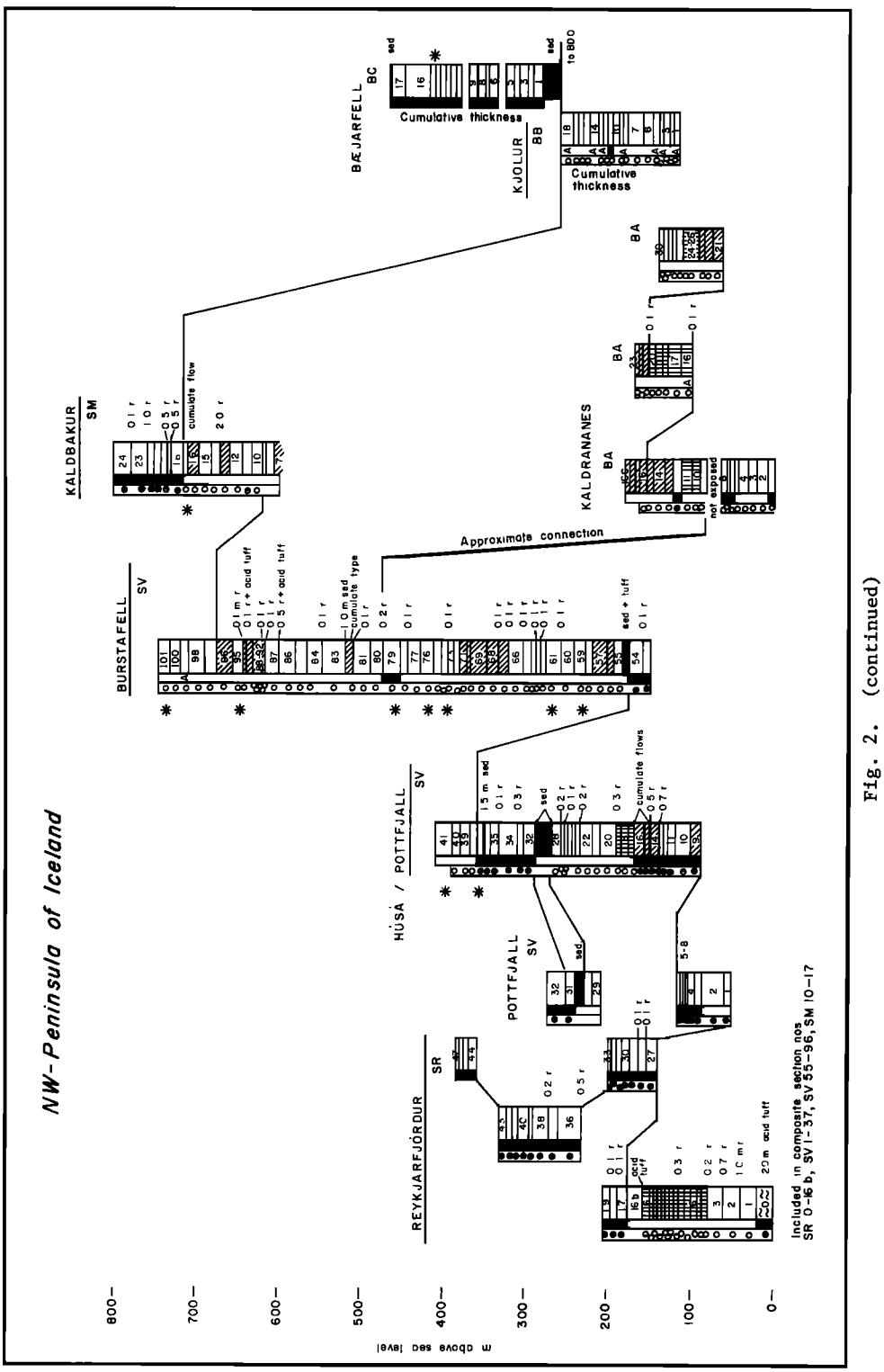


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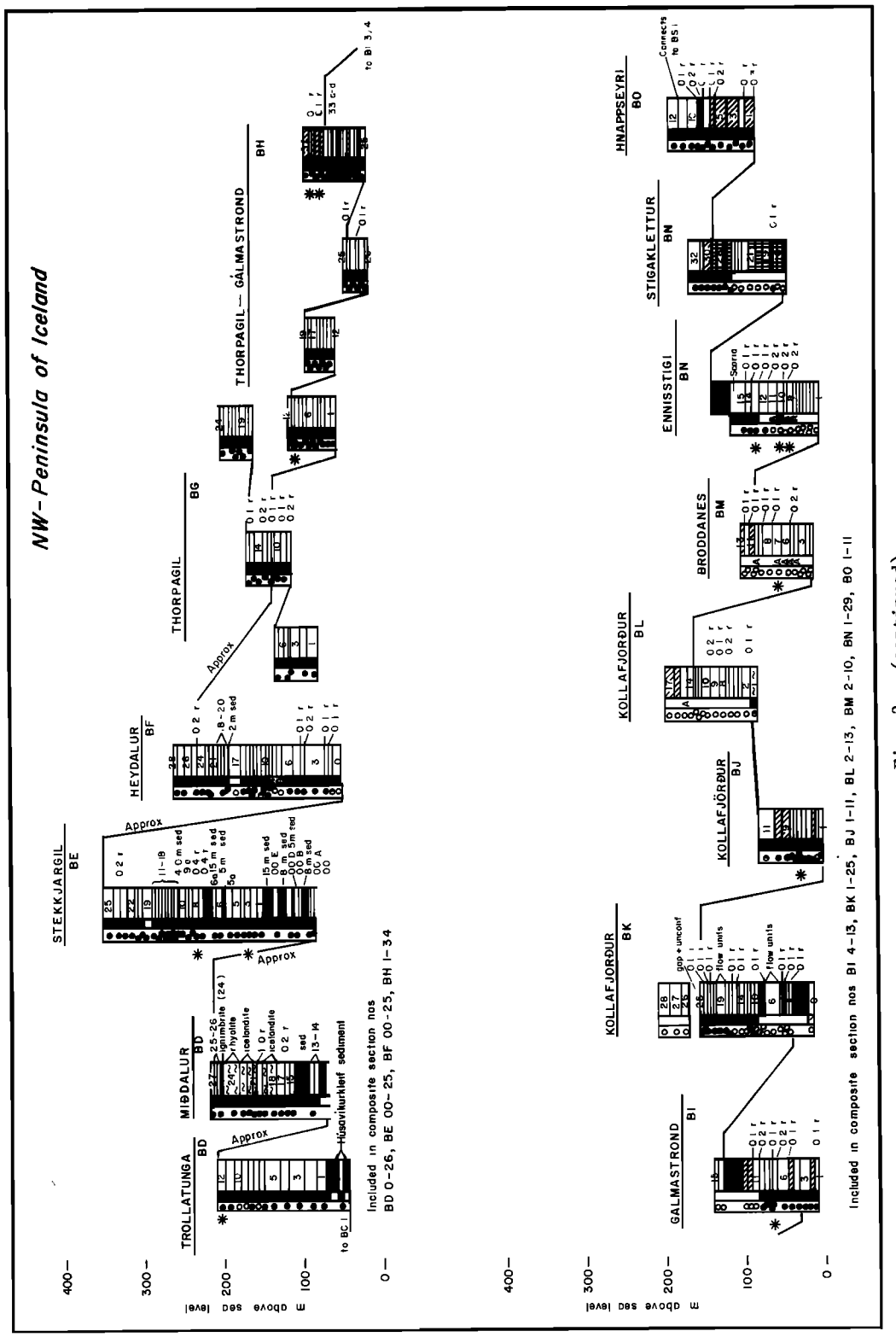


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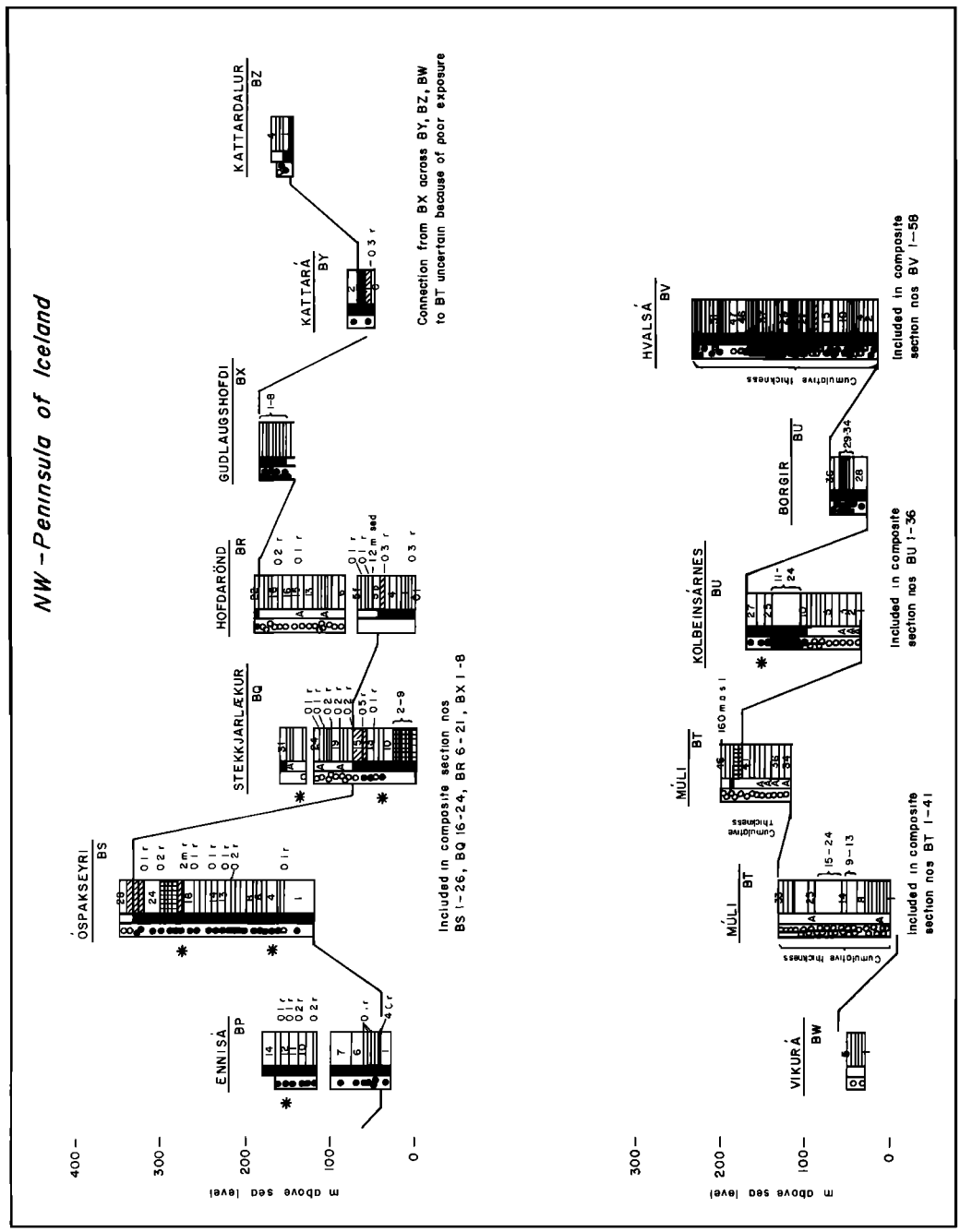


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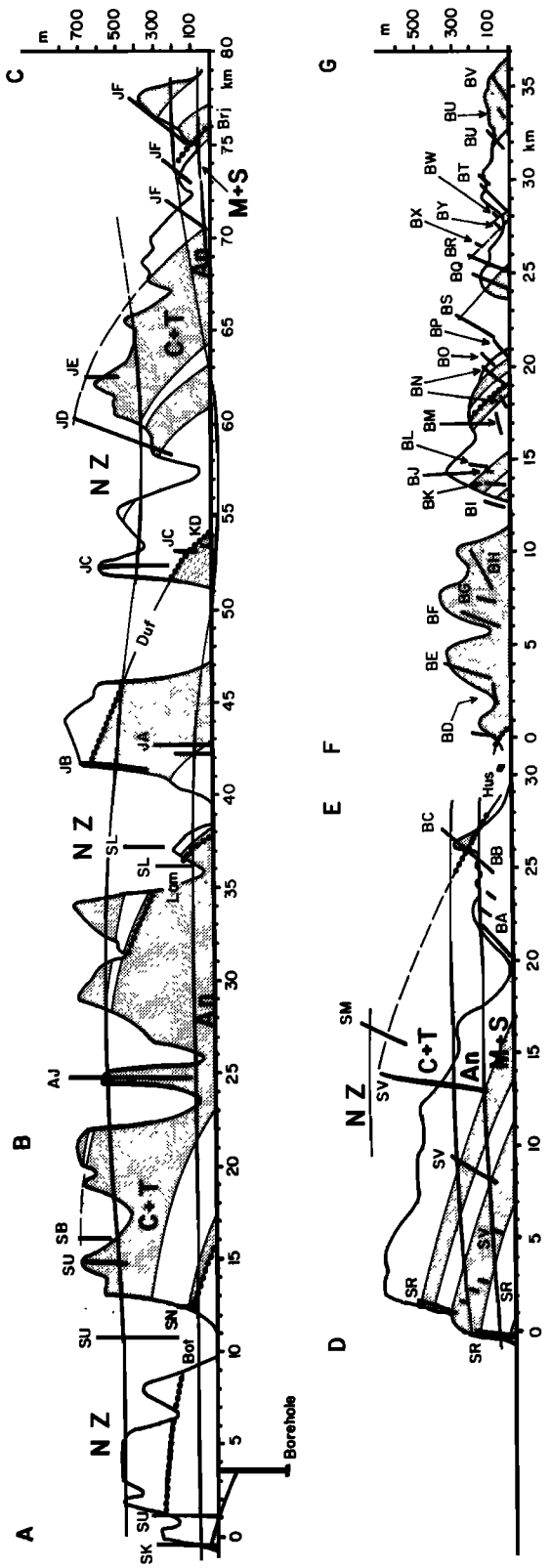


Fig. 3. Cross sections along the trace of the two main composite sections, the locations of which are shown in Figure 1. Zeolite zones, where mapped, are indicated, and location of individual profiles are shown. Note that vertical exaggeration is 10 times.

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central volcano growth to the SE (Króksfjördur) which caused an excess load and corresponding isostatic adjustment. The lava units thicken rapidly downdip, and the dips also get steeper. It appears that many of the exposed lavas belong to volcanic systems that are buried downdip. Exposed core areas of central volcanoes, however, show that some parts of the lava pile were produced near to the axis of volcanic systems. The dike density is low throughout the area, in agreement with the shallow level of erosion. Dike swarms were not recognized except in association with a central volcano beneath Reiphólsfjöll. The growth of the lava pile proceeded from NW to SE, probably in response to a more or less stationary axial rift zone located on the SE fringe of the plateau, which drifted WNW with time. The SE dip continues without interruption toward the south shore of Breidafjördur where the dips are toward the NW. This synclinal structure has been interpreted as an extinct axial rift zone [Saemundsson, 1979]. It would appear that the western composite section was entirely generated on this extinct rift zone. East of Breidafjördur the SE dip prevails to the northern extremity of the western volcanic zone (Figure 1, inset map), which therefore may be regarded as the parental axial rift zone for the eastern composite section.

Paleomagnetism

Previous Paleomagnetic Work

The first paleomagnetic survey reported from the NW peninsula of Iceland was made by Friedrich [1966], who mapped in the field the magnetic polarity of lavas in a profile of about 700 m total thickness. This profile coincides with the major part of our profile JF (Figure 2).

Kristjansson [1968] sampled 60 lavas for laboratory magnetic measurements from short profiles straddling the Botn lignite beds of our SN profile (Figure 2). He ascertained that these lavas were good material for paleomagnetic research and published a mean pole position which indicated that the Botn lignites were younger than the Eocene age that had been assigned to them previously.

Kristjansson et al. [1975] measured samples from 44 lavas in Arnarfjördur. These were also stably magnetized, belonging mostly to a normal magnetic zone underlying a hyaloclastite/lignite horizon. This zone corresponds to the normal zone sampled by us in profiles in JA to JC farther north.

Johannesson [1980] mapped magnetic polarity zones in the field in the lava pile of central western Iceland as far north as the area between Steingrimsfjördur and Króksfjördur (Figure 1). By combining this work with previous mapping in southwestern Iceland he correlated the polarity zones with the time scale of LaBrecque et al. [1977] between the Brunhes and anomaly 5 time [Johannesson, 1980, Figures 6 and 7].

Sampling

During geological mapping in the area, measurements of paleomagnetic field polarity were made on each lava using a portable flux gate magnetometer. With proper precautions [Saemundsson et al., 1980; Kristjansson, 1984] this method is quite reliable and is of considerable help in stratigraphic correlation between nearby profiles. As this polarity mapping commonly was carried out beyond the limits of the profiles subsequently sampled by coring, results from the flux gate measurements are included in Figure 2.

A total of 1261 lavas were sampled for laboratory paleomagnetic measurements in 1975-1977. They are distributed as follows between the various parts of our composite section, in the order listed in Table 1[1] and Figure 2: western composite section, profiles SK through JF: 493 lavas; southern composite section, profiles SF and SX: 83 lavas; eastern composite section, profiles SR through BX: 540 lavas; and profiles BY through BV: 145 lavas.

Generally, three cores oriented by geographic or sun sightings were collected from each flow (average of 3.1), as described previously [McDougall et al., 1977; Kristjansson et al., 1980; Saemundsson et al., 1980]. Collecting larger numbers of samples per flow is not necessary because of the very good stability of original remanence in most Icelandic lavas, the generally large secular variation, and the exceptionally large size of the present survey.

In some of the profiles BA to BV, where an unusually large number of lavas was found to yield unstable or scattered remanence directions, additional sampling (averaging two cores) was carried out in about 60 flows in 1978. The results from the additional cores have been used either to replace data from highly discordant samples or for incorporation in the calculation of overall magnetic properties along with that from the original samples.

Laboratory Measurements and Presentation of Data

Procedures follow the pattern described in our previous papers on Icelandic lava flows. The AF demagnetization was carried out at 0.01 T and 0.02 T peak fields in N. D. Watkins's laboratory, the directional results being combined using the K_{max} criterion explained by McDougall et al. [1977]. In resampled flows (see above) the more clustered of the two sets of directional data (after either 0.01 or 0.02 T for all samples used in averaging) was selected.

Values of mean tectonic tilt, which was measured during the geological mapping, are listed in Table 1C. Compilation, checking, and correction of measurement data, recalibration of intensity measurements, all work on resampled flows, and final calculations were done at the University of Iceland.

Table 1 shows tectonically corrected best field directions from all flows that were cored and the coordinates of the virtual geomagnetic pole (VGP) position for each flow. Directions are rounded off to the nearest degree, which is appropriate in view of uncertainties in geological tilt correction, and other error sources

[1]Table 1 is available with entire article on microfiche. Order from American Geophysical Union, 2000 Florida Ave., N.W., Washington, D.C. 20009.

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[Kristjansson et al., 1980]. An α_{95} is given for each flow direction. We have rejected the computed best average directions from 45 flows whose α_{95} values exceed 60°, but our estimates of their primary polarities are given in Table 1 when we feel reasonably confident of these. The major reasons for large α_{95} values [Kristjansson and McDougall, 1982] in these lavas are probably magnetic instability on one hand, and remagnetization or mistakes in sampling and measurement on the other.

Furthermore, magnetic directions in 99 lava flows whose α_{95} was between 23.5° and 60° have been excluded from our calculations of statistical properties of the geomagnetic field. One of these flows (JF 117) may have been affected by lightning, but otherwise we have no evidence of serious disturbance of remanence by lightning in our NW Iceland lava collection.

The choice of the above rejection criteria is largely arbitrary. However, results from only a small proportion of the lavas in our collection are thus excluded, namely, from 144 lavas out of a sampled total of 1261. Most conclusions regarding first-order properties of the geomagnetic field that may be drawn from our results will not be sensitive to changes in these criteria, but future survey efforts in Iceland should make the reduction of within-lava scatter a primary objective.

Table 1 also lists average volume intensities of remanence for all flows after 0.01 T AF treatment. We feel that the 0.01 T result is a fairly good estimate of the intensity of primary remanence in these; at lower fields, Brunhes age viscous remanence (VRM) has not been entirely eliminated, whereas at higher fields the intensity results may be affected by instabilities. However, 0.0125 or 0.015 T would possibly have been a better choice [cf. Dagley and Wilson, 1971]. The arithmetic mean remanence intensity in these and other Icelandic Tertiary lavas ($\alpha_{95}<23.5°$) is just over 3 A/m after 0.01 T treatment. The NRM mean intensity is about 4 A/m; from the shape of demagnetization curves for normal and reverse polarity lava samples we estimate the mean intensity of VRM to be about 0.3 A/m.

Paleomagnetic Results

Paleomagnetic data from the lavas of the NW peninsula are listed in Table 1. Results are plotted in terms of latitude of the derived VGP versus cumulative stratigraphic thickness for the western and eastern composite sections in Figures 4 and 5, respectively, with an interpreted polarity log shown on the right in each case.

Kristjansson and McDougall [1982] already have published a statistical study of some magnetic properties in the present collection of lava samples from NW Iceland, comparing these with the magnetic properties of three other lava collections of N. D. Watkins and co-workers. They demonstrated that the remanence properties of the various large paleomagnetic collections so far made in Iceland are very similar. This supports both the hypothesis that Icelandic paleomagnetic data provide a faithful record of the past geomagnetic field and the hypothesis that the overall statistical properties of the field (as averaged over several chrons) have not changed appreciably during the last 14 Ma.

It must be stressed, however, that the Neogene paleomagnetic field, when observed by measurements on Icelandic lavas, appears as a series of spot readings that have certain average properties but are otherwise only partially correlated in time (i.e., within profiles) and space (between profiles spanning the same age range). A picture of the first-order properties of this field in Iceland is emerging, but our studies are still far from providing significant information on second-order details or trends. Below, some of our results on magnetic polarity zones and mean paleomagnetic directions will be discussed. Further treatment of geomagnetic secular variation properties will be published separately [Kristjansson, 1984, and manuscript in preparation, 1984].

Polarity Zones and Transitions

In order to discuss the chronology of polarity zones in the Icelandic lava pile, one must adopt some convention to distinguish assumed reversals of the earth's main field from more temporary directional excursions. Our present definition is that the field has reversed if the virtual geomagnetic pole is found to move more than 40° across the equator in reliably magnetized lavas. This definition is very similar to that used by McDougall et al. [1977] and Saemundsson et al. [1980].

As may be expected, a number of observed short excursions in our data are excluded from polarity zone status by the above definition. These occur, for example, in the lavas SK1, SU40, SU60, SU86, JC24, JF21, SZ9-10, BF0, BF7-8, BJ2, and BN18, whereas lavas JF29-30 and BS2 marginally qualify as separate polarity zones. The one-lava polarity zones SU1, SL5, JD2, and BA12 occur within those parts of our dated sections that were subsequently designated as overlaps, and therefore they have not been included in the polarity column of Figures 4 and 5. One short polarity zone at lavas JF48A-C [Einarsson, 1963] was not sampled for laboratory measurements.

From a statistical analysis of polarity zone lengths in long dated sections through the Icelandic lava pile (including the present sections SK-JF and SR-BX), Kristjansson and McDougall [1982] infer that the geomagnetic field changed its polarity on average once every 130,000 years in the period 14-2 Ma ago. If the frequency of polarity change has been approximately constant, regions of rapid buildup of the volcanic pile should contain relatively thick polarity zones. This is indeed the case in our dated sections SK-JF (rapid buildup) and SR-BX (less rapid buildup), where average zone thicknesses (Figures 4 and 5) are 260 and 160 m, respectively. The shorter undated sections SF-SZ (to SZ79) and BY-BV, which average about 100 m per zone, may therefore have been built up even slower than SR-BX.

Mean Paleomagnetic Directions

Kristjansson and McDougall [1982, Table 3] have computed an average paleomagnetic field direction from those lava flows in the present

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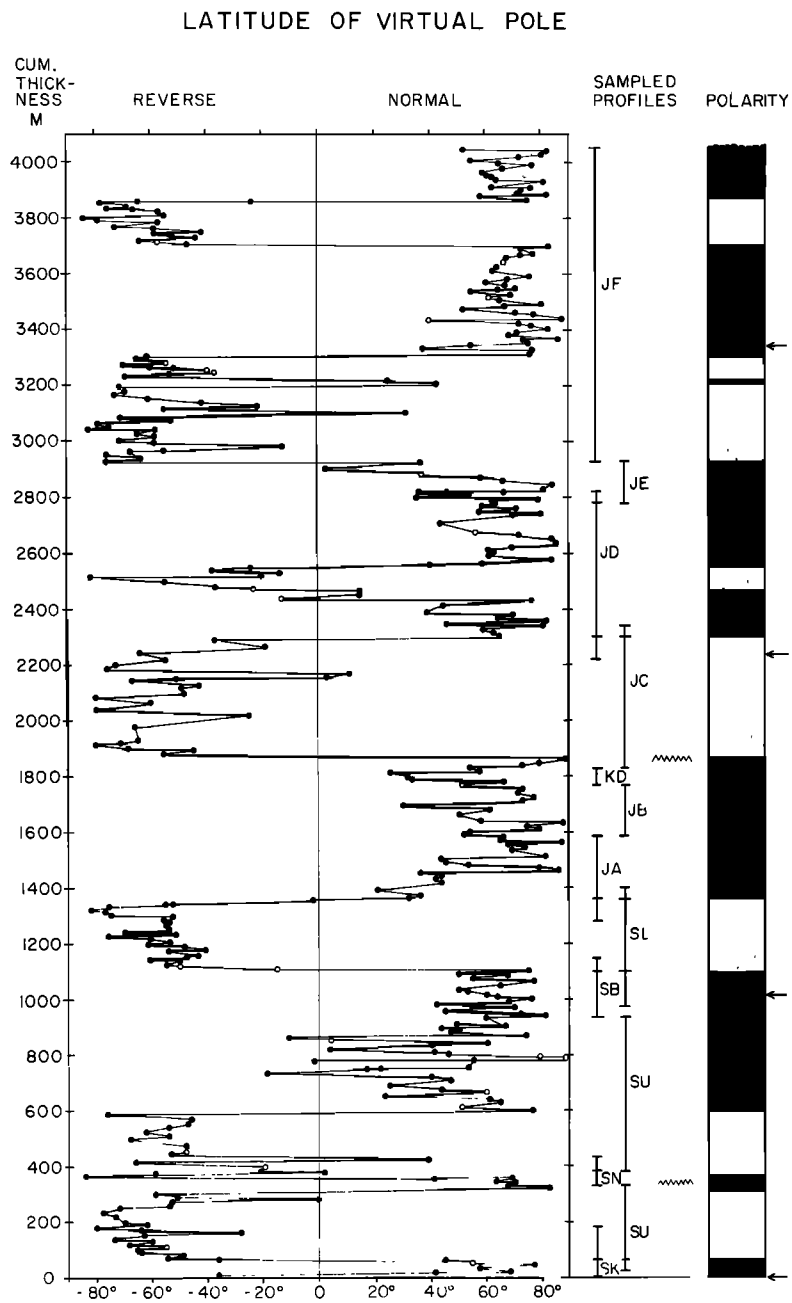


Fig. 4. Latitude of VGP derived from basalts of the western composite section plotted against cumulative stratigraphic thickness. Solid and open circles indicate flows where the α_{95} of the field direction mean is less or more than 23.5°, respectively. Short bars across vertical profile lines mark overlap with neighboring profiles, but only a single data set is plotted. Interpreted polarity log given on right, where black indicates normal polarity and white indicates reverse polarity. Arrows on the right indicate stratigraphic position of additional thin polarity zones, recorded in profiles overlapping with those of the main composite section.

collection which have an internal α_{95} value of less than 23.5°. However, in the entire collection some intervals of time may be overrepresented because of repeated sampling in overlapping profiles. In order to eliminate possible bias that may be introduced through such repeated sampling, we have also calculated a mean field from those lavas which were included in Figures 4 and 5. This mean field represents the time period between approximately 14 and 8 Ma ago.

The mean fields and their standard deviation parameters are listed in Table 2. Average pole positions are also given. It is seen that the subset of nonoverlapping lavas from the dated sections SK-JF and SR-BX yields an overall direction very similar (within 1°) to our entire data set from NW Iceland.

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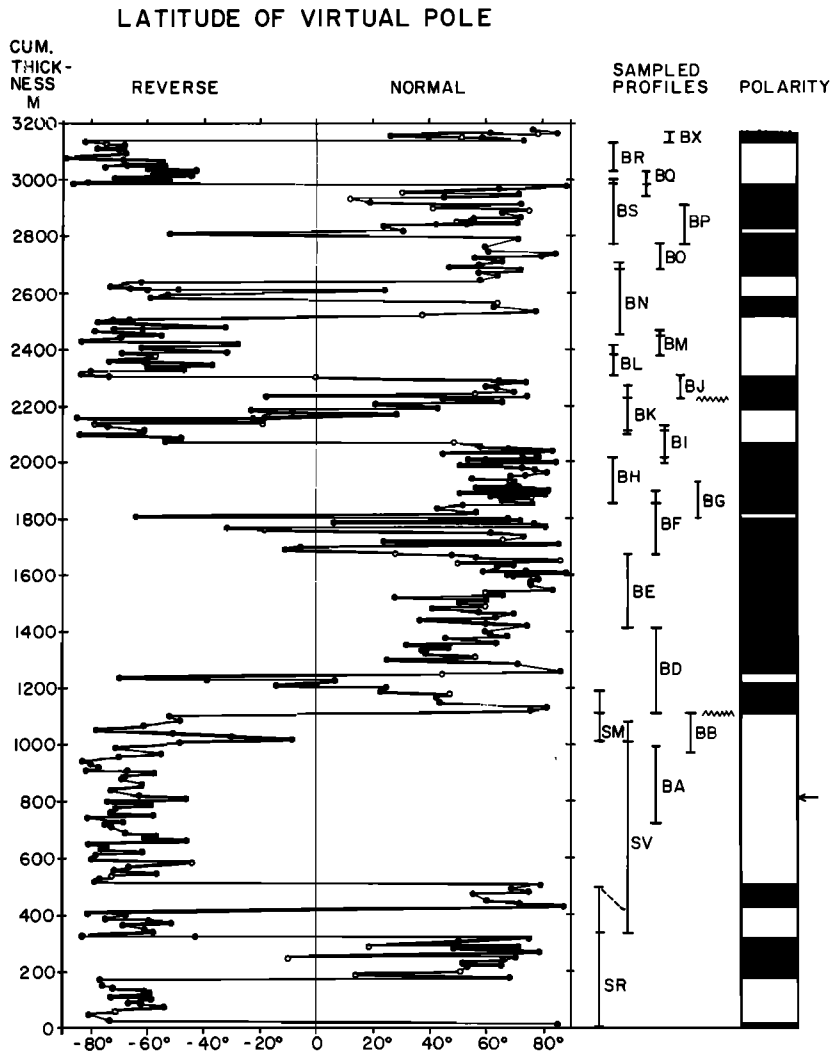


Fig. 5. Plot of latitude of VGP derived from basalt flows of the eastern composite section. Same conventions adopted as given in caption to Figure 4.

These mean directions correspond quite closely to that expected for an axial central dipole field (I ≈ 77.2°) in this area. As remarked by Kristjansson and McDougall [1982], reversely magnetized lavas yield a slightly more "far-sided" mean pole than normal lavas, whereas the latter yield a more "right-handed" pole. However, as these differences do not characterize our other paleomagnetic collections from Iceland systematically, we do not ascribe these differences to significant departures of the long-term geomagnetic field from normal/reverse symmetry.

Potassium-Argon Dating

Methods

Samples for possible dating were collected in the field by using a large hammer on the densest part of lavas that appeared to be fresh. A thin section of each sample was carefully examined under the petrographic microscope, and those that showed evidence of significant alteration or were poorly crystallized were rejected. In addition, samples exhibiting evidence of zeolitization or containing calcite or zeolite in amygdales likewise were rejected. Only those samples that were well crystallized with the main high-temperature minerals (plagioclase, pyroxene, olivine, iron oxide) essentially free of alteration were accepted for dating. Many samples contained some poorly crystallized mesostasis or intersertal glass and/or development of yellow, brown, or green mineraloid, but these were only used for dating if the total amount of such material was less than about 5%.

Each sample chosen for dating was crushed to a size of 1.0-0.5 mm, an aliquant taken and further crushed to less than 0.15 mm for K analysis. Although measurement procedures were similar to those described previously [McDougall et al., 1977], for the present study, K was determined by isotope dilution instead of flame photometry. Approximately 30 mg of the finely crushed whole rock sample was carefully weighed into a FEP beaker, a weighed amount of ^{41}K-enriched tracer added, and the material dissolved in HF and $HClO_4$ under a clean air hood. After dissolution and drying down, HCl was added; the solution was then transferred to an ion exchange column and eluted with HCl. A K cut was obtained and the isotopic composition of the K mixture was meas-

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TABLE 2. Mean Paleomagnetic Directions, NW Iceland

	N	D	I	csd	Pole A Long.	Pole A Lat.	Pole B Long.	Pole B Lat.	csd
			Including All Lavas (SK-BV)						
Normal	596	14	76.1	24.0	80	83.7	57	84.2	33.8
Reverse	521	1	74.8	25.1	151	85.8	176	89.0	33.7
All	1117	8	75.6	24.4	104	85.5	65	87.0	33.8
			Excluding Overlapping Lavas (SK-JF, SR-BX)						
Normal	404	17	76.3	23.3	73	82.6	52	82.9	32.5
Reverse	326	359	75.2	24.8	169	86.4	216	88.8	34.0
All	730	8	76.1	24.1	95	86.0	54	86.6	33.4

N, number of lavas averaged; D and I, declination and inclination of mean field; csd, circular standard deviation. Pole A is the pole position corresponding to mean field (average site coordinates 65.7, -22.5) and pole B is the average pole position from all lavas.

ured in the thermal ionization mass spectrometer described by Clement and Compston [1972]. The ^{39}K blank determined for this whole procedure averages 40 ± 12 ng, so that it accounts for less than 0.1% of the total K handled in most cases. All samples were measured in duplicate, and the pooled coefficient of variation for this series of analyses was 0.4%. The ^{41}K tracer was calibrated against KCl salt (NBS 999), and allowing for uncertainties in these calibrations, the precision of measurement of K on a single sample is of the order of 0.7%. Many of the samples were also measured by the flame photometry method employed in previous studies on Icelandic basalts, and in the majority of cases the agreement between the two methods was better than 2% relative and often was better than 1%.

Radiogenic Ar was determined by isotope dilution using a ^{38}Ar tracer as previously described, with isotopic analyses of the purified gas being undertaken in a substantially modified AEI MS10 mass spectrometer, operating on-line with a computer in which all data reduction is done immediately following the isotopic analysis. The coarsely crushed whole rock samples were employed for the Ar extractions, generally using between 10 and 20 g of material. The samples were not heated above 100°C prior to Ar extraction to minimize the possibility of isotopic fractionation of atmospheric Ar [cf. McDougall et al., 1976a]. Replicate Ar measurements were made on each sample, and with a few notable exceptions the agreement found is 2% or better (Tables 3-10). The pooled coefficient of variation for all replicates listed in the tables of data is 0.94%. In a few cases there was very poor replication of the Ar analyses with differences up to 20%. Such samples almost invariably yield anomalously high ages relative to other samples in the same sequence, so that it is thought that the problem is related to isotopic fractionation of the atmospheric Ar component in the samples during pumpdown in the vacuum system, prior to Ar extraction, with disproportionate loss of ^{36}Ar, as noted previously for a minority of Icelandic basalts [McDougall et al., 1976a]. In view of this problem, samples for which duplicate Ar analyses disagree by more than 5% have not been included in the data tables, as the measured apparent ages are considered to be unreliable.

Uncertainties in the measured ages are given at the level of one standard deviation, derived by quadratically combining errors associated with the K and Ar determinations, together with those of the tracer calibrations according to methods outlined by McDougall et al. [1969]. In the tables of results the average age given for each sample is the mean of the replicates with a quoted uncertainty either that directly calculated or that of a single determination, whichever is the greater. This is done because in those cases where the replicates agree very well the uncertainty derived for the mean age is unreasonably small, because there are finite errors associated with tracer calibrations which cannot be ignored. On average the precision of an age in terms of the physical measurements is ±1.5% (standard deviation), but, of course, this is quite independent of any geological or other effects which might have caused perturbation of the K-Ar system.

Interpretation of K-Ar ages

It has been demonstrated above that the precision of measurement of the K-Ar ages generally is better than 2%. In a few cases, especially for some profiles on the western side of the NW peninsula, age results on closely spaced samples agree to within the precision of measurement. For example, the four samples dated from different lavas in Hornataer (JE) yield K-Ar ages that are concordant, with a mean age of 12.4 ± 0.1 Ma (Table 6), and because of the consistency of these results we have great confidence that the measured age gives a precise age of eruption and cooling of the lavas. However, for most profiles the spread in measured age is greater than expected from experimental error alone, from which it is concluded that the K-Ar system for at least some samples has not behaved ideally. Loss of radiogenic Ar from slightly altered rocks causes measured ages to be less than that of eruption and cooling, and undoubtedly, this is a significant factor in producing an increased spread of apparent ages to lower values. For example, sample JD34 from Botnshestur yields an age that is 5% less than the average of results from three other lavas (JD32, JD35, JD39) from just below and just above JD34 (Table 6), so that it is inferred that this

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TABLE 3. Potassium-Argon Ages on Basalts From Sugandafjördur, West of Isafjördur, Northwest Peninsula, Iceland

Field Number	Lab Number	K, wt %	Radiogenic ^{40}Ar, 10^{-12} mol/g	100 Radiogenic ^{40}Ar / Total ^{40}Ar	Calculated Age ±1 s.d., Ma	Average Age ±1 s.d., Ma	Altitude, m	Cumulative Stratigraphic Thickness, m
				Botnsfjall				
SU100	75-190	0.5152,0.5164	12.11	42.1	13.49 ± 0.15	13.33 ± 0.21	620	895
			11.69	36.3	13.02 ± 0.16			
			12.07	41.5	13.44 ± 0.15			
			12.01	40.6	13.37 ± 0.15			
SU91	75-186	0.2517,0.2503	6.11	53.9	13.99 ± 0.15	14.01 ± 0.16	590	855
			6.20	54.9	14.19 ± 0.16			
			6.06	52.7	13.87 ± 0.15			
				Nupur				
SU70	75-183	0.4507,0.4460	10.80	38.1	13.84 ± 0.16	13.77 ± 0.16	630	810
			10.73	34.5	13.75 ± 0.17			
			10.72	35.0	13.74 ± 0.17			
SU66	75-182	0.3356,0.3362	7.86	43.0	13.44 ± 0.15	13.36 ± 0.15	605	785
			7.77	38.7	13.29 ± 0.15			
SU65	75-181	0.2247,0.2242	5.25	58.9	13.43 ± 0.14	13.49 ± 0.14	590	780
			5.30	28.2	13.56 ± 0.18			
SU60	75-179	0.2959,0.2977	7.21	49.9	13.96 ± 0.15	14.02 ± 0.15	550	745
			7.28	58.8	14.09 ± 0.15			
SU53	75-178	0.2621,0.2634	6.15	41.5	13.45 ± 0.15	13.48 ± 0.15	465	645
			6.18	42.9	13.51 ± 0.15			
SU51	75-177	0.2298,0.2282	5.28	55.5	13.24 ± 0.14	13.37 ± 0.18	430	605
			5.38	55.0	13.50 ± 0.15			
				Spillir				
SU37	75-171	0.4448,0.4383	9.85	35.1	12.81 ± 0.18	12.90 ± 0.18	340	400
			9.98	59.1	12.99 ± 0.18			
SU10	75-163	0.1190,0.1179	3.16	46.6	15.34 ± 0.17	15.32 ± 0.17	60	80
			3.16	47.0	15.30 ± 0.17			

All K-Ar ages are calculated using following constants: $\lambda_e + \lambda' = 0.581 \times 10^{-10}$ yr^{-1}, $\lambda_\beta = 4.962 \times 10^{-10}$ yr^{-1}, $^{40}K/K = 1.167 \times 10^{-4}$ mol/mol. Mean age = 13.71 ± 0.66 Ma, and mean age, excluding SU37 and SU10, = 13.60 ± 0.29 Ma.

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TABLE 4. Potassium-Argon Ages on Basalts From Burfell, 7 km West of Isafjördur, Northwest Peninsula, Iceland

Field Number	Lab Number	K, wt %	Radiogenic ^{40}Ar, 10^{-12} mol/g	100 Radiogenic ^{40}Ar / Total ^{40}Ar	Calculated Age ±1 s.d., Ma	Average Age ±1 s.d., Ma	Altitude, m	Cumulative Stratigraphic Thickness, m
				Breidadalsheidi				
P1302	76-399	0.2275,0.2266	5.608	50.1	14.25 ± 0.18	13.81 ± 0.31	538	-
			5.281	50.5	13.78 ± 0.15			
			5.337	50.6	13.52 ± 0.15			
			5.413	49.5	13.71 ± 0.15			
				Burfell				
SB17	75-202	0.1685,0.1686	3.932	50.8	13.40 ± 0.15	13.40 ± 0.15	670	1030
			3.933	40.8	13.40 ± 0.15			
SB15	75-201	0.1576,0.1585	3.787	48.2	13.76 ± 0.15	13.81 ± 0.15	650	1020
			3.811	50.0	13.85 ± 0.15			
SB13	75-200	0.1776,0.1785	4.061	54.0	13.10 ± 0.15	13.35 ± 0.23	630	1000
			4.198	39.8	13.54 ± 0.16			
			4.158	56.2	13.41 ± 0.14			
SB9	75-197	0.3418,0.3433	8.355	47.2	14.01 ± 0.15	13.97 ± 0.15	610	975
			8.314	46.7	13.94 ± 0.15			
SB5	75-195	0.5847,0.5861	13.55	48.2	13.30 ± 0.14	13.30 ± 0.15	580	945
			13.56	48.6	13.30 ± 0.15			

Burfell mean age = 13.57 ± 0.30 Ma.

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TABLE 5. Potassium-Argon Ages on Basalts From Fjallfoss and Urdafell, Arnafjördur, Northwest Peninsula, Iceland

Field Number	Lab Number	K, wt %	Radiogenic ^{40}Ar, 10^{-12} mol/g	100 Radiogenic ^{40}Ar / Total ^{40}Ar	Calculated Age ±1 s.d., Ma	Average Age ±1 s.d., Ma	Altitude, m	Cumulative Stratigraphic Thickness, m
JC30	75-252	0.2069,0.2063	4.768	54.3	13.26 ± 0.14	13.20 ± 0.14	540	2285
			4.719	55.9	13.12 ± 0.14			
			4.754	55.7	13.22 ± 0.14			
JC29B	75-251	0.2460,0.2454	5.373	51.4	12.57 ± 0.14	12.65 ± 0.14	525	2265
			5.447	51.5	12.74 ± 0.14			
JC24	75-248	0.1692,0.1703	3.852	65.1	13.04 ± 0.14	13.11 ± 0.14	430	2170
			3.902	62.6	13.20 ± 0.14			
			3.870	63.9	13.10 ± 0.14			
JC21B	75-246	0.1252,0.1253	2.836	34.7	13.01 ± 0.16	13.03 ± 0.16	410	2145
			2.848	35.4	13.06 ± 0.16			
JC21	75-245	0.1240,0.1244	2.917	38.7	13.49 ± 0.16	13.54 ± 0.16	410	2145
			2.939	39.2	13.59 ± 0.16			
JC18	75-243	0.1407,0.1415	2.910	43.7	11.85 ± 0.13	12.01 ± 0.22	360	2110
			2.987	45.7	12.16 ± 0.14			
JC15	75-242	0.1927,0.1925	4.252	61.4	12.68 ± 0.14	13.14 ± 0.38	310	2045
			4.460	62.5	13.30 ± 0.14			
			4.363	62.5	13.01 ± 0.14			
			4.546	55.5	13.56 ± 0.15			
JC11	75-240	0.2132,0.2095	4.495	57.7	12.22 ± 0.18	12.17 ± 0.18	175	1950
			4.460	61.1	12.13 ± 0.18			

Mean age = 12.86 ± 0.53 Ma, and mean age, excluding JC11 and JC18, = 13.11 ± 0.29 Ma.

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TABLE 6. Potassium-Argon Ages on Basalts From Botnshestur and Hornataer, Northwest Peninsula, Iceland

Field Number	Lab Number	K, wt %	Radiogenic ^{40}Ar, 10^{-12} mol/g	100 Radiogenic ^{40}Ar / Total ^{40}Ar	Calculated Age ±1 s.d., Ma	Average Age ±1 s.d., Ma	Altitude, m	Cumulative Stratigraphic Thickness, m
				Hornataer				
JE14	75-285	0.2075,0.2054	4.477 4.404	55.1 63.5	12.46 ± 0.14 12.26 ± 0.13	12.36 ± 0.14	685	2930
JE9	75-282	0.1972,0.1974	4.271 4.200	44.0 44.2	12.44 ± 0.14 12.23 ± 0.14	12.34 ± 0.15	610	2845
JE8	75-281	0.1895,0.1917	4.093 4.130 4.112	37.4 36.8 36.1	12.34 ± 0.15 12.45 ± 0.15 12.40 ± 0.16	12.40 ± 0.15	600	2835
JE5	75-280	0.1752,0.1748	3.815 3.778	56.4 58.4	12.53 ± 0.13 12.41 ± 0.13	12.47 ± 0.13	580	2815
				Botnshestur				
JD49	75-277	0.2344,0.2322	5.276 5.225	35.1 35.6	12.99 ± 0.18 12.87 ± 0.18	12.93 ± 0.18	700	2815
JD45	75-276	0.2109,0.2136	4.726 4.759	40.1 38.4	12.79 ± 0.17 12.88 ± 0.13	12.84 ± 0.13	670	2790
JD41	75-274	0.1584,0.1570	3.512 3.509	48.8 46.1	12.79 ± 0.17 12.79 ± 0.12	12.79 ± 0.12	635	2770
JD39	75-272	0.1538,0.1544	3.504 3.503	44.3 40.8	13.06 ± 0.17 13.06 ± 0.18	13.06 ± 0.18	620	2740
JD35	75-270	0.1953,0.1936	4.430 4.453	67.3 65.6	13.09 ± 0.24 13.15 ± 0.24	13.12 ± 0.24	580	2650
JD34	75-269	0.2177,0.2176	4.746 4.700	44.2 41.8	12.53 ± 0.17 12.41 ± 0.17	12.47 ± 0.17	570	2635
JD32	75-267	0.1526,0.1505	3.446 3.440	50.0 47.1	13.06 ± 0.18 13.04 ± 0.18	13.05 ± 0.18	560	2615

Hornataer mean age = 12.39 ± 0.06 Ma, and Botnshestur mean age = 12.89 ± 0.22 Ma.

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TABLE 7. Potassium-Argon Ages on Basalts From Brjanslaeukur, Adjacent to South Coast of Northwest Peninsula, Iceland

Field Number	Lab Number	K, wt %	Radiogenic ^{40}Ar, 10^{-12} mol/g	100 Radiogenic ^{40}Ar / Total ^{40}Ar	Calculated Age ±1 s.d., Ma	Average Age ±1 s.d., Ma	Altitude, m	Cumulative Stratigraphic Thickness, m
JF160	75-298	0.2014,0.2029	4.272	34.0	12.14 ± 0.15	11.84 ± 0.28	455	4055
			4.136	32.8	11.76 ± 0.14			
			4.082	33.1	11.61 ± 0.14			
JF156	75-297	0.2336,0.2338	4.866	38.3	11.97 ± 0.14	11.93 ± 0.14	440	4030
			4.867	39.6	11.97 ± 0.14			
			4.826	38.6	11.87 ± 0.14			
JF155	75-296	0.2456,0.2458	5.077	52.8	11.88 ± 0.13	11.72 ± 0.14	430	4025
			4.997	52.3	11.69 ± 0.13			
			4.962	52.4	11.61 ± 0.13			
JF153	75-295	0.2661,0.2670	5.551	35.2	11.97 ± 0.14	12.02 ± 0.14	420	4020
			5.600	37.5	12.07 ± 0.14			
JF151	75-294	0.2142,0.2156	4.330	37.5	11.58 ± 0.14	11.48 ± 0.14	405	4000
			4.254	35.7	11.38 ± 0.13			
JF149	75-293	0.2210,0.2219	4.477	29.5	11.62 ± 0.15	11.68 ± 0.15	395	3975
			4.522	28.1	11.73 ± 0.15			
JF140	75-289	0.1325,0.1339	2.429	46.6	10.48 ± 0.12	10.48 ± 0.12	350	3900
			2.427	43.0	10.47 ± 0.12			

Mean age, excluding JF140, = 11.78 ± 0.19 Ma.

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TABLE 8. Potassium-Argon Ages on Basalts From Kaldbakur, Burstafell and Húsá, Adjacent to East Coast of Northwest Peninsula, Iceland

Field Number	Lab Number	K, wt %	Radiogenic ^{40}Ar, 10^{-12} mol/g	100 Radiogenic ^{40}Ar / Total ^{40}Ar	Calculated Age ±1 s.d., Ma	Average Age ±1 s.d., Ma	Altitude, m	Cumulative Stratigraphic Thickness, m
				Kaldbakur				
SM17	75-339	0.1614,0.1611	3.190 3.256	25.6 25.3	11.37 ± 0.14 11.60 ± 0.15	11.49 ± 0.17	715	1105
				Burstafell				
SV101	75-337	0.1322,0.1312	2.526 2.516	48.0 47.9	11.02 ± 0.11 10.98 ± 0.11	11.00 ± 0.11	740	1070
SV95	75-335	0.1245,0.1250	2.402 2.540	52.7 52.5	11.07 ± 0.11 11.70 ± 0.12	11.38 ± 0.45	650	980
SV79	75-333	0.1585,0.1562	3.090 3.168	30.8 29.1	11.29 ± 0.16 11.57 ± 0.17	11.43 ± 0.20	470	790
SV76	75-332	0.1639,0.1648	3.347 3.411	24.3 24.8	11.70 ± 0.16 11.93 ± 0.15	11.82 ± 0.16	420	750
SV74	75-331	0.1828,0.1835	3.764 3.790	29.7 29.4	11.81 ± 0.14 11.89 ± 0.14	11.85 ± 0.14	400	730
SV61	75-329	0.2167,0.2177	3.905 4.012	46.5 64.4	10.34 ± 0.10 10.62 ± 0.10	10.48 ± 0.20	255	600
SV59	75-328	0.3858,0.3881	7.261 7.245	75.6 75.9	10.79 ± 0.10 10.75 ± 0.10	10.77 ± 0.10	230	565
				Húsá				
SV41	75-326	0.1147,0.1154	1.923 1.925	38.8 39.1	9.61 ± 0.10 9.62 ± 0.10	9.62 ± 0.10	400	545
SV37	75-325	0.1854,0.1862	2.884 2.894	44.0 41.0	8.93 ± 0.09 8.95 ± 0.10	8.94 ± 0.09	355	505

Mean age for SV101, 95, 79, 76, 74 = 11.50 ± 0.35 Ma.

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TABLE 9. Potassium-Argon Ages on Basalts From Sections Adjacent to the East Coast of Northwest Peninsula, Iceland

Field Number	Lab Number	K, wt %	Radiogenic ^{40}Ar, 10^{-12} mol/g	100 Radiogenic ^{40}Ar / Total ^{40}Ar	Calculated Age ±1 s.d., Ma	Average Age ±1 s.d., Ma	Altitude, m	Locality	Cumulative Stratigraphic Thickness, m
BM7	77-536	0.3087,0.3107	4.912 5.018	34.8 36.4	9.12 ± 0.11 9.32 ± 0.11	9.22 ± 0.14	50	Broddanes	2420
BJ6	77-533	0.2506,0.2523	4.176 4.245	27.0 28.1	9.55 ± 0.12 9.71 ± 0.13	9.63 ± 0.12	30	Kollafjördur	2255
BI7	77-531	0.4693,0.4711	6.969 7.052	37.5 41.1	8.52 ± 0.12 8.63 ± 0.12	8.58 ± 0.12	65	Galmaströnd	2050
BH37	77-529	0.1716,0.1714	2.797 2.804	22.8 24.2	9.38 ± 0.13 9.40 ± 0.17	9.39 ± 0.13	85	Galmaströnd	2040
BH35	77-528	0.1956,0.1951	3.236 3.309	21.5 22.4	9.52 ± 0.14 9.74 ± 0.16	9.63 ± 0.16	70	Galmaströnd	2025
BH11	77-527	0.1466,0.1462	2.552 2.549	31.3 33.6	10.02 ± 0.12 10.01 ± 0.16	10.02 ± 0.12	100	Galmaströnd	1910
BE8	77-514	0.2663,0.2662	4.619 4.742	34.8 35.3	9.98 ± 0.11 10.24 ± 0.12	10.11 ± 0.18	235	Stekkjargil	1560
BE3	77-513	0.1830,0.1841	3.299 3.365	31.1 25.7	10.33 ± 0.12 10.54 ± 0.14	10.44 ± 0.15	170	Stekkjargil	1495
BD12	77-510	0.4509,0.4520	8.075 8.155	27.3 44.1	10.28 ± 0.13 10.39 ± 0.11	10.34 ± 0.11	205	Tröllatunga	1270
BC15	77-503	0.4081,0.4039	7.019 7.026	38.3 38.1	9.94 ± 0.12 9.95 ± 0.12	9.95 ± 0.12	415	Baejarfell	1265

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TABLE 10. Potassium-Argon Ages on Basalts From Sections Adjacent to the East Coast of Northwest Peninsula, Iceland

Field Number	Lab Number	K, wt %	Radiogenic ^{40}Ar, 10^{-12} mol/g	100 Radiogenic ^{40}Ar / Total ^{40}Ar	Calculated Age ±1 s.d., Ma	Average Age ±1 s.d., Ma	Altitude, m	Locality	Cumulative Stratigraphic Thickness, m
BU26	77-572	0.3022,0.3026	4.152 4.362 4.109	41.0 40.7 46.8	7.90 ± 0.11 8.30 ± 0.11 7.82 ± 0.09	8.00 ± 0.26	150	Kolbeinsarnes	3660
BQ27	77-550	0.2520,0.2553	3.720 3.743 3.638	34.7 35.0 35.1	8.44 ± 0.12 8.49 ± 0.12 8.25 ± 0.11	8.39 ± 0.13	135	Stekkjarlaekur	3075
BQ11	77-549	0.2509,0.2518	3.696 3.662	27.5 30.5	8.46 ± 0.10 8.38 ± 0.10	8.41 ± 0.10	40	Stekkjarlaekur	2975
BP12	77-548	0.2346,0.2319	3.467 3.401	51.2 52.7	8.55 ± 0.10 8.39 ± 0.10	8.47 ± 0.11	155	Ennisá	2885
BS19	77-559	0.2328,0.2323	3.484 3.491	25.2 26.7	8.62 ± 0.12 8.63 ± 0.12	8.62 ± 0.12	275	Ospakseyri	2930
BS4	77-555	0.3225,0.3213	4.524 4.668	32.9 24.7	8.08 ± 0.09 8.34 ± 0.11	8.21 ± 0.18	170	Ospakseyri	2820
BN13	77-541	0.3487,0.3521	5.336 5.536	54.2 57.0	8.76 ± 0.10 9.09 ± 0.10	8.93 ± 0.23	80	Ennisstigi	2530
BN10A	77-530	0.2297,0.2325	3.542 3.609	41.0 40.0	8.81 ± 0.12 8.98 ± 0.12	8.90 ± 0.12	50	Ennisstigi	2500
BN8	77-537	0.2141,0.2136	3.298 3.333	21.9 23.2	8.87 ± 0.13 8.96 ± 0.13	8.92 ± 0.13	35	Ennisstigi	2485

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sample has lost some radiogenic Ar. High apparent ages can result from laboratory-induced loss of atmospheric ^{36}Ar relative to atmospheric ^{40}Ar in the vacuum system prior to Ar extraction, and this phenomenon also could contribute to a spread in the measured ages. Anomalously old ages also can arise if all preexisting radiogenic Ar failed to be completely outgassed from the magma when it erupted as a lava and cooled. However, the latter does not seem to be a significant problem in Icelandic lavas.

Thus there are mechanisms which can cause measured K-Ar ages to be too old or too young. In the present study the philosophy adopted is to examine the measured ages in relation to the known stratigraphic sequence, and if they are broadly consistent with this stratigraphy, they are accepted as reasonable estimates for the age of the samples. For the thick profiles in the western composite section of the peninsula, most of the samples dated were from relatively high altitudes, usually above 300 m, and from a restricted stratigraphic thickness of 100-200 m. In these cases it is often possible to calculate an average age for the restricted thickness sampled, and if a measured age is more than 4% removed from the mean age for that section, it is regarded as anomalous. The profiles on the east side of the peninsula generally are much thinner, and those samples considered suitable for age measurement are rather more evenly spaced throughout the sequence, so calculation of average ages usually is not possible. Thus consistency of the ages with the stratigraphy is the main criterion used to assess the reliability of the measured ages for the eastern composite section.

Discussion of Age Data

The K-Ar age results are listed in Tables 3-10 in stratigraphic order. Cumulative stratigraphic thickness above the base for each sample is given separately for the composite sections on the western and eastern side of the peninsula as firm correlation between the two areas, about 90 km apart, has not been possible. The samples show a wide range of K contents, from 0.11 to 0.58%, with the majority having values lying between 0.1 and 0.3%. The radiogenic ^{40}Ar relative to the total ^{40}Ar in the gas extracted from the samples ranges from 21 to 75%, averaging about 40%.

The age data are interpreted to indicate that the complete sequence of flood basalt lavas in the NW peninsula, comprising about 7 km in thickness, ranges in age from approximately 14 Ma near the base (above lava SN1) to about 8 Ma near the top of the composite section, covering an interval of about 6 Ma.

Age data from the western composite section are plotted versus cumulative stratigraphic thickness in Figure 6. The monotonic decrease in measured age upward through the succession provides considerable confidence that these ages give a good estimate for the time of eruption and cooling of the lavas. Nevertheless, because the measured ages from samples in close stratigraphic proximity show a spread beyond that which can be accounted for by experimental error and because of the presence of several anomalous results, brief discussion of data from individual profiles is necessary.

The stratigraphically oldest sequence in the NW peninsula examined in this study in profiles at Spillir, Núpur, and Botnsfjall (Figure 2) has proved to be somewhat difficult to date precisely. Measured ages (Table 3) range from 15.3 ± 0.2 Ma (SU10) to 12.9 ± 0.2 Ma (SU37) on samples collected from the lower part of the sequence. Ages from eight samples higher in the sequence show a much more limited spread in apparent age from 13.3 to 14.0 Ma but without any correlation with stratigraphic position. The mean age for these eight samples of 13.6 ± 0.3 Ma probably is the best estimate that can be derived from these data for this part of the sequence. The significantly younger apparent age of 12.9 ± 0.2 Ma for sample SU37 may be explained by loss of radiogenic Ar. The old apparent age of 15.3 ± 0.2 Ma for sample SU10 could be realistic, as there is strong evidence from the geology for a significant break in the stratigraphic succession below lava SU34 in the Spillir profile or SN1 in the Botn profile. Unfortunately, no other samples from this part of the profile were found to be fresh enough for dating to enable this conclusion to be checked.

The lavas in Búrfell (SB) likewise yield a spread in measured age uncorrelated with the stratigraphy (Table 4). The mean age of 13.5 ± 0.2 Ma calculated from the five results is accepted as a reasonable estimate for the age of this sequence; note that it is indistinguishable from the age derived for the underlying Núpur profile. Sample P1302 from Breidadalsheidi, thought to be approximately at the same stratigraphic level as the Búrfell section, yields an apparent age consistent with that view. The measured age of 13.8 ± 0.3 Ma agrees well with the preferred age of 14.1 ± 0.2 Ma for Breidadalsheidi lavas given by Albertsson and Einarsson [1982].

Eight basalt samples from the profiles in Fjallfoss and Urdarfjall (JC) give apparent ages ranging from 12.0 ± 0.2 Ma to 13.5 ± 0.2 Ma (Table 5), although excluding the low ages obtained for samples JC11 and JC18, on the assumption of loss of radiogenic Ar, provides a satisfactory and stratigraphically consistent average of 13.1 ± 0.3 Ma.

The profile in Botnshestur (JD) yields apparent ages ranging from 12.5 ± 0.2 to 13.1 ± 0.2 Ma based upon results from seven samples (Table 6) collected over a stratigraphic thickness of 110 m. The mean age of 12.9 ± 0.2 Ma is changed only slightly to 13.0 ± 0.1 Ma if the low result from sample JD34 is excluded. The mean age is indistinguishable from that found for the underlying profile JC.

The five samples from Hornataer (JE) give concordant results (Table 6) and provide a mean age of 12.4 ± 0.1 Ma, regarded as a very secure age for this profile.

Only samples from the upper 100 m of the profile in Brjánslaekur (JF) were suitable for dating; results from six of these (Table 7) yield a concordant set averaging 11.8 ± 0.2 Ma, if the low apparent age for sample JF140 is excluded.

As noted previously, the plot of measured age versus stratigraphic thickness in Figure 6 for

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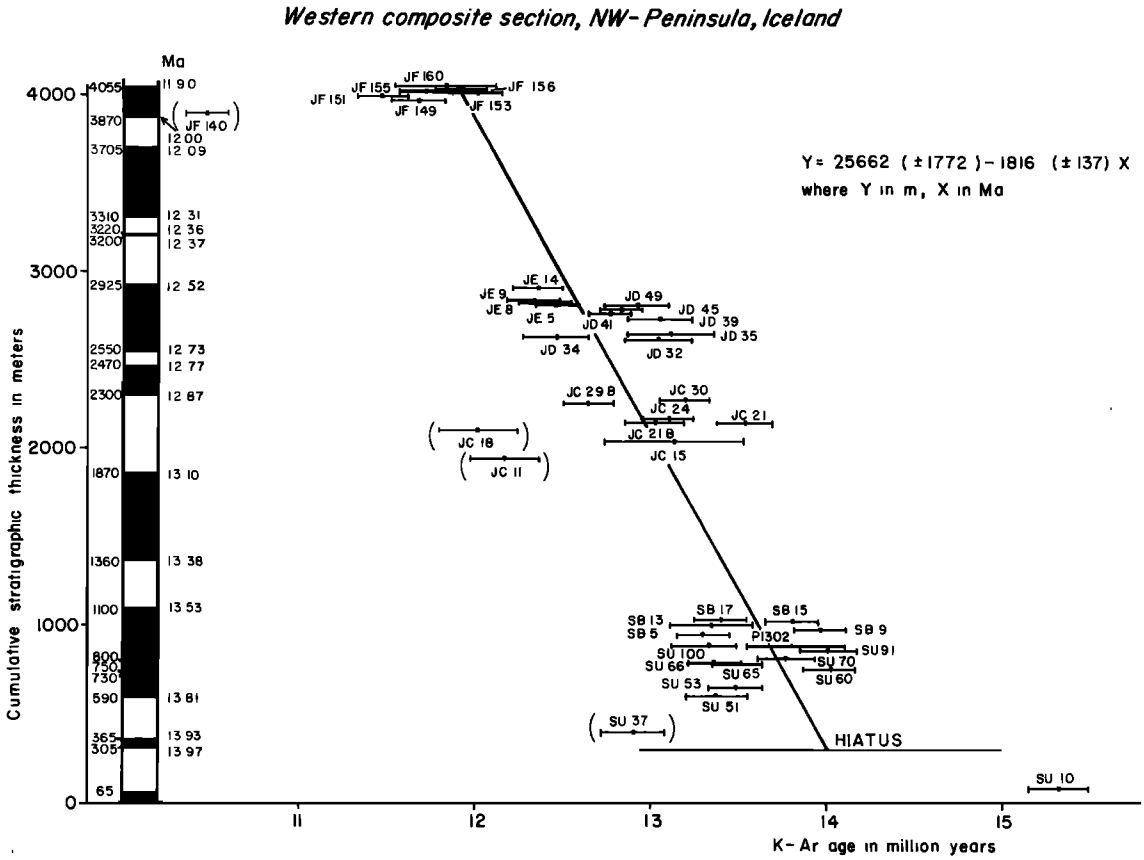


Fig. 6. Plot of measured K-Ar age against cumulative stratigraphic thickness for the western composite section. Error bars for K-Ar ages are one standard deviation. Best fit regression line through data is shown; age results in parentheses not used in regression analysis. Polarity log for the sequence is given adjacent to thickness axis, where black indicates normal polarity and white indicates reverse polarity. Ages for polarity interval boundaries derived from the regression.

the composite section in the western part of the peninsula shows the general consistency of the results. A linear relation between age and stratigraphic thickness is evident, similar to that found in our earlier studies of Icelandic flood basalt sequences. Regression of these data using the procedure of York [1969], which allows for errors in both parameters, provides the following best fit straight line:

$$Y = 25662\ (\pm 1772) - 1816\ (\pm 137)\ X$$

where Y is the cumulative stratigraphic thickness in meters and X is the age in million years. The uncertainties quoted are standard errors, which are about 6.9 and 7.6% for the intercept on the stratigraphic thickness axis and for the slope, respectively. In the regression analysis the obviously aberrant ages (SU10, SU37, JC11, JC18, and JF140) have been omitted, so that a total of 36 data points was used. For the regression analysis an error of 2.2% was assigned to the ages, and the uncertainty for the cumulative stratigraphic thickness parameter arbitrarily was given a value of 5%. With these parameters the line fits to within experimental error (MSWD=1.0).

The good alignment of the data indicates that a model of uniform growth of the lava pile fits the results quite well. The slope of the regression line provides an estimate of 1816 m/Ma for the rate of growth of the lava pile, and this is within the range of values obtained in our earlier studies of other lava sequences in Iceland. Although the data indicate an overall uniform growth of the lava sequence, we cannot exclude the possibility that short-term variations in rates of eruption of lavas have occurred. In this respect, note also that the age data are on samples from restricted parts of the composite section with two separate segments of the sequence, each about 1000 m thick, with no age measurements.

Thus for the western part of the NW peninsula above lava SN1 we conclude that volcanism extended from 14.0 to 11.9 Ma ago using the regression data. These data indicate that the eruption of the 3750-m lava succession extended over an interval of about 2.1 Ma in the middle Miocene. As some 420 lavas have been mapped in the succession, the average time between successive lavas is calculated to be 5000 years.

Because of the generally rather low topography on the east side of the NW peninsula, the composite section was constructed from numerous short stratigraphic profiles. Despite the fact that lavas collected for dating were from deeper in the lava pile, compared with those obtained from the western composite section, a reasonable

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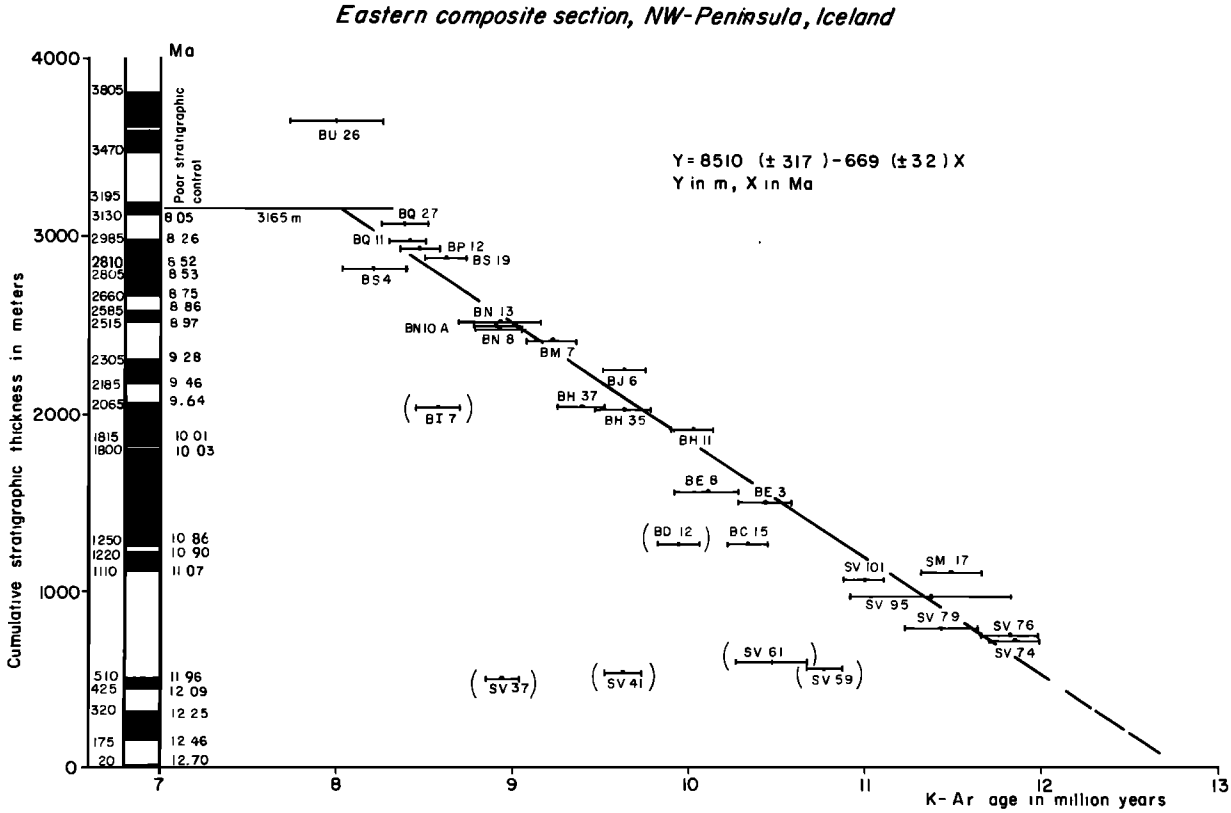


Fig. 7. Plot of measured K-Ar age against cumulative stratigraphic thickness for the eastern composite section. Best fit regression line through data is shown; age results in parentheses not used in regression analysis. Ages for polarity interval boundaries indicated on polarity log are derived from the regression.

number of samples were thought to be suitable for dating, and these are distributed fairly uniformly through the sequence.

The age data (Tables 8-10) are plotted against cumulative stratigraphic thickness above the base of the composite section in Figure 7, and as for the western composite section, there is good consistency of the measured ages with the stratigraphy. The overall trend of younging upward is accepted as strong evidence that the results provide a satisfactory measure of age of eruption and cooling of the lavas. However, five of the 26 measured ages clearly are anomalously young (SV37, SV41, SV59, SV61, BI7). These samples probably have lost radiogenic Ar, and the results are excluded from further discussion. A model of uniform growth of the lava pile fits the bulk of the data; regression analysis yields the following best fit straight line:

$$Y = 8510\ (\pm 317) - 669\ (\pm 32)\ X$$

where, as previously, Y is the cumulative stratigraphic thickness in meters and X is the age in millions of years. The intercept has an uncertainty of 3.7% and the slope an uncertainty of 4.7%, where these values are standard errors expressed as a percentage. This line is calculated using an error of 5% for the stratigraphic thickness parameter and an assigned error of 1.8% or as calculated from the replicates, which ever is the greater, for the K-Ar ages. These errors are chosen somewhat arbitrarily in order to ensure that the line fits to within error (MSWD=1.0). The regression yields a rate of growth for the lava pile of 669 m/Ma, less than half the value found for the western composite section. Also from the regression we estimate that the 3100-m composite section on the eastern side of the peninsula was erupted from 12.7 to 8.0 Ma ago, over an interval of 4.7 Ma. These results are consistent with the stratigraphic evidence that the eastern profiles mainly are above those on the western side of the peninsula but that there is some overlap. A total of 395 lavas was identified up through profile BX in the eastern composite section, so that on average the time interval between successive lava flows is 11,900 years.

Geomagnetic Polarity Time Scale

In Figures 6 and 7, magnetic polarity logs for the western and eastern composite sections are plotted adjacent to the stratigraphic thickness axis. The regressions provide the means by which an age for individual polarity changes can be obtained, and these ages are shown next to the polarity logs. Problems are encountered in assigning formal uncertainties to these derived ages because of lack of knowledge as to the ratio of the variance of the errors in the two parameters of the regression. The case that leads to a maximum estimate of the confidence interval is when all error is assumed to be associated with the stratigraphic thickness parameter.

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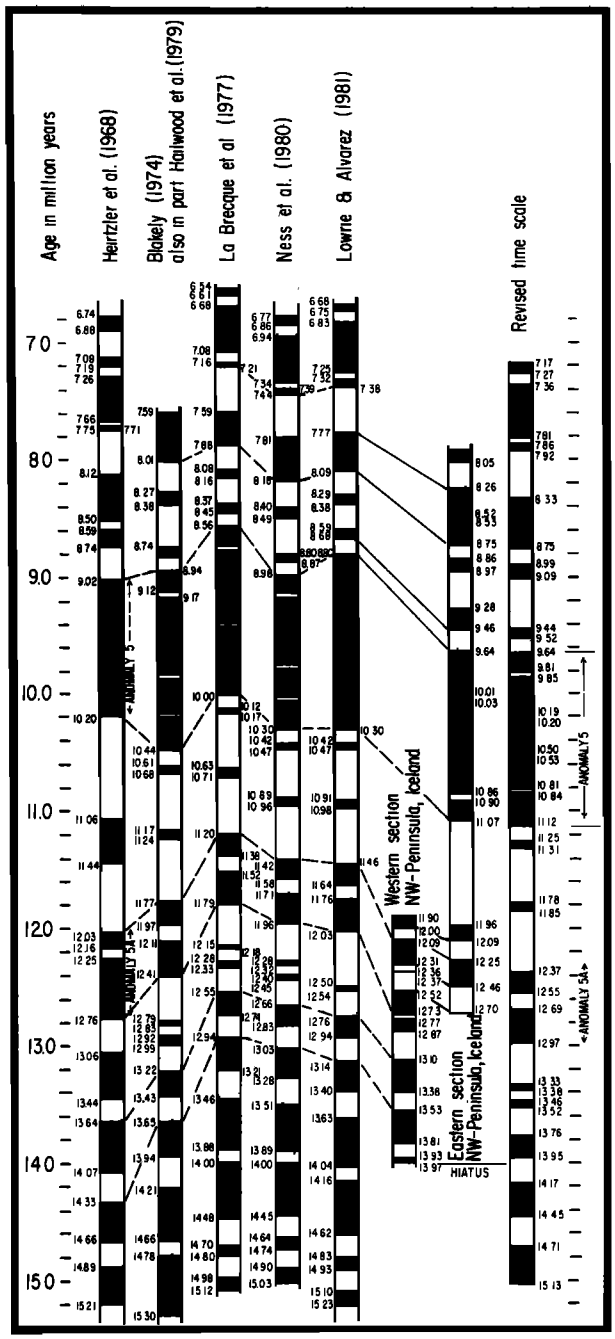


Fig. 8. Polarity time scales from various authors for the interval 7-15 Ma ago, adjusted where necessary to conform with current ^{40}K decay constants. Black represents normal polarity, white reverse polarity. Polarity logs for western and eastern composite sections of NW peninsula, Iceland, are shown to right. Suggested correlations indicated by tie lines. On extreme right is a derived polarity time scale based upon an age of 9.64 Ma for younger boundary of anomaly 5, an age of 3.40 Ma for the Gauss/Gilbert chron boundary, and acceptance of the marine magnetic anomaly pattern.

Calculations on this basis yield uncertainties for the ages for particular polarity boundaries in the eastern composite section ranging from 2.5 to 3.6% at the level of one standard deviation. As these are maximum uncertainties, this emphasizes that statistically the ages derived from the regression for the polarity interval boundaries are well constrained. For the western composite section the uncertainties are about twice as great.

In Figure 8 the polarity logs for the lava sequences in NW Iceland are plotted together with a number of polarity time scales derived from marine magnetic anomaly data. All the time scales shown are calculated, or where necessary recalculated, to conform with the ^{40}K decay constants currently in use [Steiger and Jäger, 1977]. Each of the time scales developed from marine magnetic anomaly data is based upon slightly different assumptions (see Ness et al. [1980] for summary). But each time scale essentially is calibrated by reference to the geomagnetic polarity time scale for the last 4 Ma or so as determined by direct measurement of subaerial lavas of known polarity [McDougall, 1979; Mankinen and Dalrymple, 1979] and, in particular, an age of about 3.4 Ma for the Gauss/Gilbert chron boundary. Extrapolation to older ages depends on assumptions as to seafloor spreading rates and/or the age of particular calibration points. Thus the differences between the various time scales in Figure 8 are because of the different assumptions or control ages used in their construction and emphasize the need to attempt direct calibration of the polarity time scale, one of the aims of the present study.

From Figure 8 it will be seen that an unambiguous correlation can be made between the polarity sequence determined in the eastern composite section in NW Iceland and the marine magnetic anomaly pattern but that it is not at all obvious how the polarity pattern from the western composite section should be correlated. These results illustrate the general problem encountered in correlating polarity data from land-based sections with the marine anomaly record because of the lack of uniqueness of particular polarity intervals. Correlations normally can only be effected by identification of a distinctive pattern of polarity with time.

In the case of the composite section from the east coast of the NW peninsula, the long, dominantly normal polarity interval in the middle of the sequence is confidently correlated with anomaly 5 of the marine magnetic profiles. This interpretation is strongly supported by the good match of the polarity logs above this level (Figure 8). The younger boundary of the anomaly 5 normal polarity interval lies at 2065 m in the composite eastern section, with estimated age of 9.64 Ma. However, there is some ambiguity as to the location of the older boundary of anomaly 5 in the sections. Two interpretations are possible: the boundary could lie at 1250 m with age of 10.86 Ma or at 1110 m with estimated age of 11.07 Ma (Figures 5 and 7). This is because the reverse polarity zone between 1220 and 1250 m in the section is represented by four lavas, two of which have low-latitude virtual geomagnetic poles and two have clearly reverse polarity (Figure 5), and thus might well represent the earliest of the short reverse polarity subchrons that occur within anomaly 5. A further complication is that the base of this dominantly normal polarity interval in the eastern composite section coincides with

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the base of profiles BC and BD, where a sedimentary bed occurs, possibly indicating a significant hiatus in the volcanism (Figure 2). On balance we favor the interpretation that the older limit of anomaly 5 is located at 1110 m with an age of about 11.07 Ma.

A correlation of the lower one third of the polarity log for the eastern composite section with the marine magnetic record also is shown in Figure 8 in which the stratigraphically lowest complete normal polarity interval (175-320 m, Figure 5) is correlated with the younger normal polarity interval of marine magnetic anomaly 5A. For this interpretation to be correct we have to postulate that a normal polarity interval about 0.07 Ma in duration, occurring below the base of anomaly 5, is not represented in the lava sequence as sampled adjacent to the east coast of the NW peninsula. On the basis of the average time between successive lavas in this sequence, calculated previously, such an interval should be represented by about six lava flows.

The polarity log derived from the western composite section in the NW peninsula, in contrast, shows only a rough correspondence with any of the marine magnetic logs (Figure 8). Whether this is because of short-term irregularity in lava eruption rates, masked in the overall linear relation between age and cumulative stratigraphic thickness, because additional polarity intervals have been detected in the lava sequence but not resolved in the marine magnetic record, or because there are unrecognized problems in tying the stratigraphy of the lava profiles into a continuous section, we are unable to ascertain. On the basis of the derived ages for the eastern and western composite sections and the polarity logs, assuming that the ages in both lava sequences are equally reliable, we believe that a good correlation between them is possible, as shown in Figure 8. As we have already effected a good correlation between the eastern composite section and the marine magnetic anomaly polarity logs, we can propose a tentative correlation for the western composite section with the marine magnetic record as indicated in Figure 8. However, because of the rather poor match of the polarity logs we have reservations about this interpretation.

The good correlation of the polarity pattern in the eastern composite section with the marine magnetic record enables a direct evaluation of the polarity time scale over the interval from 8.0 to 12.5 Ma ago. For comparative purposes we shall use the polarity time scale derived from the marine magnetic anomaly data by Ness et al. [1980]. For the 15 polarity interval boundaries that can be correlated with the Ness et al. [1980] time scale we find that the results from the eastern sequence in NW Iceland indicate a consistently older age of between 0.45 and 0.83 Ma (Figure 8), averaging 0.59 ± 0.12 Ma, emphasizing the homogeneity and comparability of the two data sets. This change in calibration of the polarity time scale to older ages by up to 8% is quite large, but if our data are accepted at face value, such a revision is required.

In the Ness et al. [1980] polarity time scale the duration of anomaly 5 is given as 1.32 Ma, and from the eastern composite section in NW Iceland we obtain a duration of 1.43 Ma, again attesting to the comparability of the two types of data. Of considerable significance is the fact that an important calibration point in the Ness et al. [1980] time scale is an age of 10.30 Ma for the older limit of anomaly 5, adjusted from an age of 10.55 Ma that otherwise would have been used by them from assessment of the earlier time scales of Heirtzler et al. [1968] and Blakely [1974]. Their calibration point was derived from an estimate of age for the older boundary of anomaly 5 made by McDougall et al. [1976b], recalculated to conform with the new K decay constants and based upon K-Ar age and magnetic polarity data from a thick sequence of lavas in eastern Iceland. An additional study of the same kind made on a thick lava sequence in northern Iceland by Saemundsson et al. [1980] provided yet another estimate of 10.47 Ma for the older limit of anomaly 5. Thus from Icelandic lava successions we have three independent estimates for the age of the older boundary of anomaly 5 of 10.30, 10.47, and possibly 11.07 Ma, a spread exceeding 7%. These three derived ages are sufficiently different from one another that factors other than uncertainties associated with the measurements or with the regressions almost certainly need to be invoked. One possible explanation is that there has been variable but systematic loss of radiogenic Ar from these low K, slightly altered basalts, in which case, estimates yielding the oldest ages would be regarded as more nearly correct. On this basis a conservative, and possibly systematically young, estimate for the age of the older boundary of anomaly 5 is the average of the three values obtained from the study of Icelandic lava sequences. The derived age of 10.61 ± 0.40 Ma, it is salutory to note, is very similar to the recalculated age of 10.44 Ma assigned to the same level by Blakely [1974] and to the age initially calculated by Ness et al. [1980] for the base of anomaly 5. A less conservative approach is to accept the new data from NW Iceland at face value and use it to derive a new time scale.

Part of such a time scale is given in Figure 8 for the interval from 3.40 Ma ago, the estimated age for the Gauss/Gilbert chron boundary [Mankinen and Dalrymple, 1979; McDougall, 1979], to 15 Ma ago, the older limit of our results from the NW peninsula of Iceland. It is now widely recognized that the marine magnetic anomaly data provide an excellent time-integrated history of the polarity of the geomagnetic field. Therefore we have adjusted the time scale of Ness et al. [1980] to an age of 9.64 Ma for the younger boundary of anomaly 5 as derived from the results of the present study, calculating ages for all other boundaries relative to this calibration point and the age accepted for the Gauss/Gilbert boundary. Thus the relative durations of polarity intervals interpreted from the marine magnetic record are preserved in the new time scale. It should be emphasized that choice of the younger boundary of anomaly 5 as a major calibration point is arbitrary; ages derived from the regression for other polarity boundaries could equally well have been used for this purpose, resulting in slightly different time scales. We did not employ our estimate of age for the older boundary of anomaly 5 for this purpose, although it has been commonly so used, because of ambiguity as to where it lies

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in the eastern composite section of the NW peninsula.

Comparisons between the calculated polarity time scale (Figure 8) and that derived from the eastern composite section (Figure 8) show that differences in age ranging up to 0.24 Ma are found for individual polarity interval boundaries and that the average difference is 0.00 ± 0.12 Ma based on comparison between the ages of 17 polarity boundaries. This is regarded as very satisfactory when it is recalled that the relative lengths of polarity intervals in the two sets of data were obtained quite independently of one another, being brought together by acceptance of a common age for the younger boundary of anomaly 5.

Overall, for the interval 8-15 Ma, our proposed time scale is closer to the polarity time scales of Blakely [1974] and Hailwood et al. [1979] and, for the older part, the original extrapolated time scale of Heirtzler et al. [1968] than to the time scales proposed by LaBrecque et al. [1977], Ness et al. [1980], and Lowrie and Alvarez [1981]. For the latter two time scales we stress again that they were adjusted to younger ages in part because of our earlier estimate of 10.30 Ma for the age of the older boundary of anomaly 5, based upon data from eastern Iceland. However, our time scale still yields ages about 5% older than those given by Blakely [1974].

As reemphasized by Ness et al. [1980], any time scale of this kind is subject to change as more reliable calibration points become available, and we regard the present attempt to produce an improved time scale in a similar light. It is unlikely that the approach used here can be extended to more than 15 Ma ago because of the absence of lava sequences greater than this age in Iceland. And we are not aware of older subaerial lava sequences, elsewhere in the world, in which a fairly uniform rate of eruption of lavas may have occurred over an extended period of time that might be suitable for this purpose. As recognized by most workers, calibration of the older parts of the polarity time scale is most readily achieved indirectly through correlation of magnetic polarity patterns in sedimentary sequences with the marine magnetic record and by reference to the numerical time scale determined by isotopic dating of suitable rocks whose positions in the relative geological time scale are known more or less precisely from biostratigraphic data (see Berggren [1981] for a recent review of this approach).

Summary and Conclusions

The NW peninsula of Iceland is built mainly of basaltic lavas that dip regionally gently to the SE. It is inferred that the lavas were erupted from an axial rift zone, striking in a NE direction, located on the SE margin of the growing and NW spreading lava plateau. The aggregate stratigraphic thickness exceeds 7 km. Two composite sections set up adjacent to the western and eastern coasts of the NW peninsula, respectively, were sampled in detail for paleomagnetic measurements. K-Ar dating on carefully selected lavas from throughout the composite sections shows that the sequence was erupted mainly between 14 and 8 Ma ago in the middle and late Miocene. A plot of measured age versus stratigraphic thickness yields good straight lines for the western and eastern composite sections but of different slope. From these relations, average rates of growth of the lava pile of 1820 and 670 m/Ma are found for the western and eastern composite sections respectively; these values are within the range previously recorded by us from lava sequences elsewhere in Iceland. The average time between eruption of lavas is calculated to be 5000 and 12,000 years for the western and eastern sections, respectively.

The two composite sections are about 90 km apart along strike, with a zone of flexure between them. Although the eastern composite section is predominantly younger than that in the west, the evidence shows that there is some overlap in time. The major difference in eruption rates between the two sections may reflect a sudden change in magma production and eruption as lavas with time or may indicate a marked change along the rift zone in the volcanic production rate. We are unable to distinguish between these possibilities.

The magnetostratigraphy together with the K-Ar dating enable comparison with the geomagnetic polarity time scale interpreted from marine magnetic anomaly data. For the western composite section in the NW peninsula it was not possible to derive an unequivocal correlation of the observed magnetozones and the marine magnetic anomaly pattern at least in part because of a lack of distinctiveness of the polarity pattern. For the eastern composite section, however, a good match of the magnetozones and the marine magnetic anomaly pattern was achieved owing to the distinctiveness of the polarity sequence, especially the long normal polarity interval associated with anomaly 5 of the marine magnetic record. We obtain an age of 9.64 Ma for the younger boundary of anomaly 5 from the eastern composite section magnetostratigraphy and geochronology. This has been combined with an age of 3.40 Ma for the Gauss/Gilbert chron boundary to calibrate the marine magnetic anomaly pattern and produce a revised polarity time scale back to 15 Ma ago. The new time scale is much closer to those derived by Heirtzler et al. [1968] and Blakely [1974] than to more recent versions but yields ages for individual boundaries that are about 5% greater than in the Blakely [1974] time scale. From the NW peninsula data we estimate that anomaly 5 time extended from 11.1 to 9.6 Ma ago. As previously noted, a number of the more recent polarity time scales use the age of the older boundary of anomaly 5 as a calibration point, commonly employing our earlier estimate [McDougall et al., 1976b] of age of 10.3 Ma from eastern Iceland for this boundary. Clearly, if the new results from NW Iceland are accepted at face value, considerable adjustment of these time scales to older ages is required. The differing results obtained from measurements on Icelandic lava sequences for the age of a particular magnetozone boundary highlight the difficulties associated with K-Ar dating of low K, slightly altered basalts and suggest that measured ages commonly may be systematically young owing to some loss of radiogenic Ar. Overall the consistency of the results from the eastern composite

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section of the NW peninsula, Iceland, is such that we believe that these are among the best data currently available from Icelandic lava sequences.

Finally, the location of lavas assigned to anomaly 5 in the NW peninsula of Iceland lie essentially on strike with anomaly 5 as mapped in the seafloor both north and south of the peninsula [Talwani and Eldholm, 1977; Vogt et al., 1980]. This structural coherence implies similarity in the genetic processes, although much larger volumes of magma clearly must have been erupted, compared with the oceanic crust, to build Iceland well above sea level.

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